Def $(z_n)_{n\geq p}$ - șir de nr complexe. $L \in \mathbb{C}$

$\lim_{n\to\infty} z_n = L$ dacă

$(\forall)\ \varepsilon > 0$, $(\exists)\ n_\varepsilon \geq p \in \mathbb{N}$ a.î. $(\forall)\ n \geq n_\varepsilon$ $|z_n - L| < \varepsilon$

Dacă există un astfel de L, atunci el este unic
(z_n este convergent la L)

$(z_n)_{n\geq p}$ s.n. convergent dacă există un astfel de L cu proprietatea de mai sus.

$|z_n - L| \in \mathbb{R}$

Obs. $z_n \to L \iff \underbrace{|z_n - L|}_{\in \mathbb{R}} \to 0$

$||z_n - L| - 0| < \varepsilon$

Obs! $z_n \overset{\in \mathbb{C}}{\to} 0 \iff |z_n| \to 0$

Fie $v \in \mathbb{C}$, $|v| < 1$

$z_n = v^n$ $\quad |z_n| = |v|^n \to 0 \Rightarrow z_n \to 0$

$0 \leq |v| < 1$

Prop. Dacă $|v| < 1$, $v \in \mathbb{C}$ $\Rightarrow v^n \to 0$

$z_n = x_n + i y_n \in \mathbb{C}$, $x_n, y_n \in \mathbb{R}$

$L = a + ib$, $a, b \in \mathbb{R}$

Pp. $z_n \to L \iff |z_n - L| \to 0 \iff |x_n - a + i(y_n - b)| \to 0 \iff$

$\iff \sqrt{(x_n - a)^2 + (y_n - b)^2} \longrightarrow 0$

$|x_n - a| = \sqrt{(x_n-a)^2} \leq \sqrt{(x_n-a)^2+(y_n-b)^2} \to 0$

$|y_n - b| = \sqrt{(y_n-b)^2} \leq$

$x_n \to a$, $y_n \to b$

Reciproc:

$x_n \to a$ si $y_n \to b$

$|z_n - L| = \sqrt{\underbrace{(x_n-a)^2}_{\downarrow 0}+\underbrace{(y_n-b^2)}_{\downarrow 0}} \to 0 \Rightarrow z_n \to L$

(P) Fie $z_n = x_n + i y_n$, $x_n, y_n \in \mathbb{R}$, $n \geq 2$

1) $(z_n)_n$ este convergent $\Leftrightarrow$ $(x_n)_n$ si $(y_n)_n$ sunt amb. convergente

2) $a, b \in \mathbb{R}$

$z_n \to a+ib \Leftrightarrow \begin{cases} x_n \to a \\ y_n \to b \end{cases}$

$z_n = x_n + iy_n$

Prop. Dacă $z_n \xrightarrow{\in \mathbb{C}} L_1 \in \mathbb{C}$ si $v_n \xrightarrow{\in \mathbb{C}} L_2 \in \mathbb{C}$, atunci:

1) $(\forall)\ \alpha, \beta \in \mathbb{C} \Rightarrow \alpha z_n + \beta v_n \to \alpha L_1 + \beta L_2$

2) $z_n \cdot v_n \to L_1 \cdot L_2$ etc.

Obs Fie $(z_n)_{n \geq p}$ sir de nr. complexe

Pp. că $(\exists)\ v \in \mathbb{C}$ si $q \geq p$, $q \in \mathbb{N}$ a.î.

$(\forall)\ n \geq q$, $z_n = v$ Atunci $z_n \to v$.

Ex: Convergența - să se studieze
(Calc. limita)

a) $z_n = \frac{5n^2+1}{n^2+2} + i\left(1+\frac{1}{n}\right)^n$

b) $v_n = \frac{n+5}{2n+1} + i(-1)^n$

a) $\frac{5n^2+1}{n^2+2} + i\left(1+\frac{1}{n}\right)^n \longrightarrow 5 + i \cdot e$

b) $v_n = \frac{n+5}{2n+1} + i(-1)^n$
$\quad\quad\quad

Dacă seria $\sum_{n\geq q} z_n$ este convergentă:

se not. $\sum_{n=q}^{\infty} z_n \overset{def}{=} \lim_{n\to\infty} S_n$ și se numește suma seriei.

Obs $\sum_{n\geq q} z_n$, $\sum_{n\geq p} z_n$

$p > q$, $p \in \mathbb{N}$

$n \geq p$

$S_n = z_q + \dots + z_{p-1} + z_p + z_{p+1} + \dots + z_n$

C | T_n

↓ Șirul sumelor parțiale asociat seriei $\sum_{n\geq q} z_n$

→ Șirul sumelor parțiale asociate seriei $\sum_{n\geq p} z_n$

$S_n = C + T_n$, $(\forall) n \geq p$

Obs $\sum_{n\geq q} z_p$ este convergentă $\iff$ $\sum_{n\geq p} z_n$ este convergentă

Dacă sunt convergente avem

$\sum_{n=q}^{\infty} z_n = z_q + z_{q+1} + \dots + z_{p-1} + \sum_{n=p}^{\infty} z_n$

Proprietăți

1) Dacă $\sum_{n\geq q} z_n$ și $\sum_{n\geq q} v_n$ sunt convergente, at. $(\forall) \alpha, \beta \in \mathbb{C} \Rightarrow$

$\Rightarrow \sum_{n\geq q} (\alpha z_n + \beta v_n)$ este convergentă și $\sum_{n=q}^{\infty} (\alpha z_n + \beta v_n) =$

$= \alpha \sum_{n=q}^{\infty} z_n + \beta \sum_{n=q}^{\infty} v_n$

2) Dacă $\sum_{n\geq 0} z_n$ este convergentă și $(\forall)\ \alpha \in \mathbb{C}$

$\Rightarrow \sum_{n\geq 0} \alpha z_n$ este convergentă și $\sum_{n=0}^{\infty} \alpha z_n =$

3) Dacă $\sum_{n} z_n$ este divergentă și $\alpha \in \mathbb{C}, \alpha \neq 0 \Rightarrow$

$\Rightarrow \sum_{n} \alpha z_n$ este div.

(Dem pp. prin absurd că $\sum_n (\alpha z_n)$ ar fi convergentă $\Rightarrow$

$\Rightarrow \sum_n \frac{1}{\alpha}(\alpha z_n)$ este conv. $\sum_n z_n$ conv. este fals)

4) Dacă $\sum_n z_n$ este conv și $\sum_n v_n$ este div și $\alpha, \beta \in \mathbb{C}, \beta \neq 0 \Rightarrow$

$\Rightarrow \sum_n (\alpha z_n + \beta v_n)$ este div.

$\sum_n (z_n \pm v_n)$ este div

$v_n = z^n$
$\sum_{n\geq 0} v^n$

Seria geometrică

Teorema Fie $z \in \mathbb{C}$ $\sum_{n\geq 0} z^n$ (Șirul sumelor parțiale: $S_n = 1 + z + z^2 + \dots + z^n$)

1) Dacă $|z|<1$, atunci $\sum_{n\geq 0} z^n$ este conv. și $\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}$

2) Dacă $|z| \geq 1$, atunci $\sum_{n\geq 0} z^n$ este divergentă

Dem: 1) $|z|<1$

$S_n = 1 + z + z^2 \dots + z^n = \frac{z^{n+1}-1}{z-1} = \frac{1-z^{n+1}}{1-z} = \frac{1-(z^n)\cdot z}{1-z} \to$ ($z^n \to 0$)

$\Rightarrow S_n \to \frac{1}{1-z}$

Teoremă: (Criteriul Necesar de convergență)

$z_n \in \mathbb{C}$ Dacă $\sum_n z_n$ este conv. atunci $z_n \to 0$

2) $|z| \geq 1 \Rightarrow |z|^n \geq 1 \Leftrightarrow |z^n| \geq 1, (\forall) n \geq 1 \Rightarrow \lim_{n\to\infty} |z^n| \geq 1$ (1)

$\sum_{n\geq 0} v_n$, $v_n = z^n$

Pp. că $\sum_{n\geq 0} v_n$ este conv. $\Rightarrow v_n \to 0$, $\lim_{n\to\infty} z^n = 0 \Rightarrow$

$\Rightarrow \lim_{n\to\infty} |z^n| = 0$ (2) $\Rightarrow 0 \geq 1 \Rightarrow$ Fals

Dem Teorema

$\sum_{n\geq p} z_n$

$S_n = \underbrace{z_p + z_{p+1} + \ldots + z_{n-1}}_{S_{n-1}} + z_n$, $(\forall)_{n \geq p+1}$

$z_n = S_n - S_{n-1}$, $(\forall) n \geq p+1$

$\sum_n z_n$ conv, $(\exists) S \in \mathbb{C}$ a.î. $S_n \to S \Rightarrow S_{n-1} \to S$

$\Rightarrow \underbrace{S_n - S_{n-1}}_{z_n} \to S - S = 0 \Big/ \Rightarrow z_n \to 0$

Obs $\sum_{n\geq \varrho} z_n$, $\varrho \in \mathbb{N}$

$\sum_{n\geq \varrho} z_n$ este conv $\Leftrightarrow \sum_{k \geq \varrho+p} z_{k-p}$ este conv.

$(n = k-p,\ k = n+p \geq \varrho+p)$

Dacă sunt convergente atunci avem $\sum_{n=\varrho}^{\infty} z_n = \sum_{n=\varrho+p}^{\infty} z_{n-p}$

Ex: Studiați conv. seriilor și suma lor (dacă sunt conv)

a) $\sum_{n\geq 0} \frac{1}{2^n}$; b) $\sum_{n\geq 3} \frac{1}{2^n}$

a) $\sum_{n\geq 0} \left(\frac{1}{2}\right)^n$, $z = \frac{1}{2}$, $|z| = \frac{1}{2} < 1 \Leftrightarrow \sum_{n\geq 0} \frac{1}{2^n}$ este conv. și (serie geometrică)

$$\sum_{n=0}^{\infty} \frac{1}{2^n} = \frac{1}{1-\frac{1}{2}} = \frac{1}{\frac{1}{2}} = 2$$

b) $\sum_{n\geq 3} \frac{1}{2^n}$

$\sum_{n\geq 0} \frac{1}{2^n}$ este conv $\Rightarrow \sum_{n\geq 3} \frac{1}{2^n}$ conv

$$\sum_{n=3}^{\infty} \frac{1}{2^n} = \underbrace{\sum_{n=0}^{\infty} \frac{1}{2^n}}_{=2} - 1 - \frac{1}{2} - \frac{1}{2^2} = 1 - \frac{3}{4} = \frac{1}{4}$$

Abel.

$$\sum_{n=3}^{\infty} \frac{1}{2^n} = \frac{1}{2^3} \sum_{n=3}^{\infty} \frac{1}{2^{n-3}} = \frac{1}{2^3} \underbrace{\sum_{k=0}^{\infty} \frac{1}{2^k}}_{=2} = \frac{1}{2^3} \cdot 2 = \frac{1}{4}$$

$k = n-3 \geq 0$, $n \geq 3$

Ex Să se $\sum_{n\geq 1} \frac{n+1}{n+5}$ este div $\Leftarrow \frac{n+1}{n+5} \to 1 \neq 0$ (rezultă)

Obs Dacă $z_n \not\to 0$, atunci $\sum_n z_n$ este div

Ex $\sum_{n\geq 1} \frac{(-1)^n \cdot n+1}{n+5}$ este div $\Leftarrow$ nu există $\lim_{n\to\infty} \frac{(-1)^n \cdot n+1}{n+5}$

Serii cu termeni pozitivi

$\sum_{n \geq q} x_n$, $x_n \geq 0$, $x_n \in \mathbb{R}$, $(\forall) n \geq q$

$S_n = x_q + x_{q+1} + \dots + x_n$

$S_{n+1} = \underbrace{x_q + x_{q+1} + \dots + x_n}_{S_n} + x_{n+1}$

$S_{n+1} = S_n + \underbrace{x_{n+1}}_{\geq 0} \geq S_n$

$S_{n+1} \geq S_n$, $(\forall) n \geq q$

S_n - șir crescător

a) $x_N + \dots + x_n \geq y_N + \dots + y_n \to \infty$

$\sum_{n \geq N} y_n$ div

$\Rightarrow x_N + \dots x_{nn} \to \infty \Rightarrow$

$\Rightarrow \sum_n x_n$ div.

Obs $\sum_{n \geq q} x_n$ este conv. $\Leftrightarrow S_n$ este mărginit superior

$x_n \geq 0$ $(\forall) n \geq q$ $\sum_{n \geq q} x_n$ este div $\Leftrightarrow S_n \to \infty$

(T) $\sum_{n \geq q} x_n$, $\sum_{n \geq q} y_n$, Fie $N \geq q$, $N \in \mathbb{N}$

Pp. că.

1) $0 \leq x_n \leq y_n$, $(\forall) n \geq N$

Dacă $\sum_n y_n$ este conv $\Rightarrow \sum_n x_n$ este conv

2) Pp. că $x_n \geq y_n \geq 0$, $(\forall) n \in \mathbb{N}$

Dacă $\sum_n y_n$ este div $\Rightarrow \sum_n x_n$ este div

Dem

1) $x_N + x_{N+1} + \dots + x_n \leq y_N + y_{N+1} + \dots + y_n \leq M$

$\sum_{n \geq N} y_n$ conv $\Rightarrow y_N + \dots + y_n \leq M$ $(\exists M)$ $\Rightarrow (x_N + \dots + x_n)_n$ convergent $\Rightarrow$

$\Rightarrow \sum_{n \geq} x_n$ convergent

(T) $x_n \geq 0$, $(\forall) n \geq q$; $y_n > 0$, $(\forall) n \geq q$

Pp. că $(\exists) \lim_{n\to\infty} \frac{x_n}{y_n} \overset{not}{=} L \in [0, +\infty)$

1) Dacă $L \neq \infty$ și $\sum_n y_n$ este conv $\Rightarrow \sum_n x_n$ este conv.

2) Dacă $L \neq 0$ și $\sum_n y_n$ este div $\Rightarrow \sum_n x_n$ este div

3) Dacă $L \neq \infty$ și $L \neq 0$ atunci $\sum_n x_n$ și $\sum_n y_n$ au aceeași natură

Dem

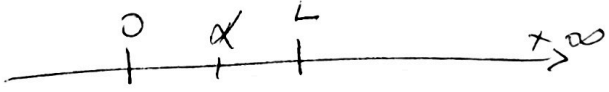
1) $\left(\frac{x_n}{y_n}\right)_n$ este conv $\Rightarrow \left(\frac{x_n}{y_n}\right)_n$ este mărginit $\Rightarrow$

$\Rightarrow (\exists) M > 0$ $\frac{x_n}{y_n} \leq M$, $(\forall) n \geq q$

$0 \leq x_n \leq M \cdot y_n$

$\sum_n y_n$ conv $\Rightarrow \sum_n M y_n$ conv $\Big| \Rightarrow \sum_n x_n$ conv

2) $\frac{x_n}{y_n} \to L > 0$



Fie $\alpha \in (0, L)$

$(\exists) N \in \mathbb{N}$, $N \geq q$ a.î. $(\forall) n \geq N$ $\frac{x_n}{y_n} \in (\alpha, +\infty)$

$\frac{x_n}{y_n} > \alpha$, $x_n > \alpha y_n > 0$

$\sum_n y_n$ div $\Rightarrow \sum_n \alpha y_n$ div $\Big| \Rightarrow \sum_n x_n$ div

---

Prop: $\sum_{n\geq 1} \frac{1}{n^\alpha}$, $\alpha \in \mathbb{R}$

1) Dacă $\alpha > 1 \Rightarrow \sum_{n\geq 1} \frac{1}{n^\alpha}$ este conv

2) Dacă $\alpha \leq 1 \Rightarrow \sum_{n\geq 1} \frac{1}{n^\alpha}$ este div

$y_n = \frac{1}{n^\alpha}$

Corolar Fie $x_n \geq 0$, $(\forall) n \geq q$. Fie $\alpha \in \mathbb{R}$.

Pp. că $(\exists) \lim_{n\to\infty} \frac{x_n}{\frac{1}{n^\alpha}} \overset{not.}{=} L \in [0, +\infty]$

1) Dacă $L \neq \infty$ și $\alpha > 1 \Rightarrow \sum_n x_n$ conv.

2) Dacă $L \neq 0$ și $\alpha \leq 1 \Rightarrow \sum_n x_n$ e div.

3) Dacă $L \neq \infty$, și $L \neq 0$ at. $\sum_n x_n$ și $\sum_n \frac{1}{n^\alpha}$ au aceeași natură

T Criteriul radicalului

$x_n \geq 0$, $(\forall) n \geq q$

Pp. că $(\exists) \lim_{n\to\infty} \sqrt[n]{x_n} \overset{not.}{=} L \in [0, +\infty]$

1) Dacă $L < 1 \Rightarrow \sum_n x_n$ este conv

2) Dacă $L > 1 \Rightarrow x_n \to \infty$, deci $\sum_n x_n$ este div.

Prop $x_n > 0$, $(\forall) n \geq q$

Pp. că $(\exists) \lim_{n\to\infty} \frac{x_{n+1}}{x_n} \overset{not}{=} L \in [0, +\infty]$

At. $(\exists) \lim_{n\to\infty} \sqrt[n]{x_n}$ și $\lim_{n\to\infty} \sqrt[n]{x_n} = L$

Crit. rap: $x_n > 0$, $(\forall) n \geq q$

Pp. că $\frac{x_{n+1}}{x_n} \to L \in [0, +\infty]$

1) Dacă $L < 1 \Rightarrow \sum_n x_n$ este conv

2) Dacă $L > 1 \Rightarrow x_n \to \infty$, $\sum_n x_n$ este div

(T) Crit. RAABE-DUHAMEL

Fie $x_n > 0$, $(\forall) n \geq 0$

Pp. că $(\exists) \lim_{n\to\infty} n\left(\frac{x_n}{x_{n+1}} - 1\right) = l \in \overline{\mathbb{R}}$

1) Dacă $l > 1 \Rightarrow \sum_n x_n$ este conv

2) Dacă $l < 1 \Rightarrow \sum_n x_n$ este div

(T) Criteriului integral

Fie $f : [p, \infty) \to [0, +\infty)$ descrescător $p \in \mathbb{N}$

Al. seria $\sum_{n\geq p} f(n)$ este convergentă (resp. div) $\Leftrightarrow$ șirul $\left(\int_p^n f(x)\,dx\right)_{n\geq p}$ este convergent (resp. div)

Ex. a) $\sum_{n\geq 1} \frac{n!}{10^n}$; b) $\sum_{n\geq 1} \left(1 - \frac{1}{n}\right)^{n^2}$ c) $\sum_{n\geq 1} \frac{n! \cdot e^n}{1\cdot 3\cdot 5 \cdot \ldots \cdot (2n+1)}$

d) $\sum_{n\geq 1} \frac{\sqrt[4]{n+2} - \sqrt[4]{n-1}}{\sqrt[3]{n}}$

a) $\sum_{n\geq 1} \frac{n!}{10^n}$ $x_n > 0$

$\frac{x_{n+1}}{x_n} = \frac{(n+1)!}{10^{n+1}} \cdot \frac{10^n}{n!} = \frac{n+1}{10} \to \infty$ div.

$\Rightarrow \sum_{n\geq 1} x_n$ div

b) $\sum_{n\geq 1}\left(1-\frac{1}{n}\right)^{n^2}$ $\quad n\geq 1 \Rightarrow \frac{1}{n}\leq 1$

$\| \; x_n \geq 0$

$\sqrt[n]{x_n} = \left(1-\frac{1}{n}\right)^n = \left(\left(1+\frac{-1}{n}\right)^{-n}\right)^{-1} \to e^{-1} = \frac{1}{e} < 1 \Rightarrow$

$\Rightarrow \sum_n x_n$ este conv.

(dacă $y_n \neq 0 \to 0 \Rightarrow (1+y_n)^{\frac{1}{y_n}} \to e$)

c) $\sum_{n\geq 1} x_n$, $x_n = \frac{n!\cdot c^n}{1\cdot 3\cdot 5\cdot \ldots\cdot (2n+1)} > 0$

$\frac{x_{n+1}}{x_n} = \frac{(n+1)!\cdot c^{n+1}}{1\cdot 3\cdot \ldots\cdot (2n+1)(2n+3)} \cdot \frac{1\cdot 3\cdot \ldots\cdot (2n+1)}{n!\cdot c^n} =$

$= \frac{c(n+1)}{2n+3} \to \frac{c}{2} = L < 1 \Leftrightarrow \frac{c}{2} < 1$

1) Dacă $0 < c < 2 \Rightarrow$
$\Rightarrow L < 1 \Rightarrow \sum_n x_n$ conv.

2) Dacă $c > 2 \Rightarrow L > 1 \Rightarrow$
$\Rightarrow \sum_n x_n$ div.

3) Dacă $c = 2$ $\quad \frac{x_{n+1}}{x_n} = \frac{c(n+1)}{2n+3} = \frac{2(n+1)}{2n+3}$

$n\left(\frac{x_n}{x_{n+1}} - 1\right) = n\left(\frac{2n+3}{2n+2} - 1\right)$

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o1) $\sum_{n\geq 1} x_n$, $x_n = \dfrac{\sqrt[4]{n+2} - \sqrt[4]{n-1}}{\sqrt[3]{n}} > 0 \quad (\forall) n \geq 1$

$$\frac{x_n}{\frac{1}{n^\alpha}} = n^\alpha \cdot \frac{\left(\sqrt[4]{n}\sqrt[4]{1+\frac{2}{n}} - \sqrt[4]{n}\sqrt[4]{1-\frac{1}{n}}\right)}{\sqrt[3]{n}} = n^{\alpha+\frac{1}{4}-\frac{1}{3}} \cdot \left(\sqrt[4]{1+\frac{2}{n}} - \sqrt[4]{1-\frac{1}{n}}\right)$$

$$= n^{\alpha-\frac{1}{12}} \cdot \frac{\sqrt{1+\frac{2}{n}} - \sqrt{1-\frac{1}{n}}}{\sqrt[4]{1+\frac{2}{n}} + \sqrt[4]{1-\frac{1}{n}}} = n^{\alpha-\frac{1}{12}} \cdot \frac{}{\left(\sqrt{1+\frac{2}{n}} + \sqrt{1-\frac{1}{n}}\right)\left(\sqrt[4]{1+\frac{2}{n}} + \sqrt[4]{1-\frac{1}{n}}\right)}$$

$$\frac{x_n}{\frac{1}{n^\alpha}} = \frac{3 \cdot n^{\alpha-\frac{1}{12}-1}}{\left(\sqrt{1+\frac{2}{n}} + \sqrt{1-\frac{1}{n}}\right)\left(\sqrt[4]{1+\frac{2}{n}} + \sqrt[4]{1-\frac{1}{n}}\right)}$$

$$\alpha = \frac{13}{12} > 1$$

$$\frac{x_n}{\frac{1}{n^{\frac{13}{12}}}} = \frac{3}{\left(\sqrt{1+\frac{2}{n}} + \sqrt{1-\frac{1}{n}}\right)\left(\sqrt[4]{1+\frac{2}{n}} + \sqrt[4]{1-\frac{1}{n}}\right)} = \frac{3}{2 \cdot 2} \neq \infty \neq 0$$

$\sum_n x_n$ și $\sum \frac{1}{n^{\frac{13}{12}}}$ au același nat $\Rightarrow$

$\Rightarrow$ ambele conv.

Inegalități utile

1) $(\forall) x \in \mathbb{R}\setminus\{0\}$, $e^x > x+1$

2) $(\forall) x > -1$, $x \neq 0$, avem $x > \ln(x+1)$

3) $(\forall) x > 0$, $x > \sin x$

4) $(\forall) x > 0$, $x > \operatorname{arctg} x$

5) $(\forall) x \in \left(0, \frac{\pi}{2}\right)$, $x < \operatorname{tg} x$

Teoremă

Fie $A \subseteq \mathbb{R}$, $A \neq \emptyset$, fie $\alpha \in \bar{\mathbb{R}}$ un punct de acumulare al mulțimii A. Fie $L \in \bar{\mathbb{R}}$. Fie $f: A \to \bar{\mathbb{R}}$ o funcție. Atunci:

$\lim\limits_{x \to \alpha} f(x) = L \iff$ oricare ar fi $(x_n)_n$ un șir de elemente din $A\setminus\{\alpha\}$, cu $x_n \to \alpha$, avem $f(x_n) \to L$

Def α se num. punct de acumulare al mulțimii A dacă există cel puțin un șir de elemente din $A\setminus\{\alpha\}$, $(x_n)_n$, cu $x_n \to \alpha$.

$x_n = f\left(\frac{1}{n}\right)$, $f: (0, a) \longrightarrow [0, \infty)$, $a > 0$

$\frac{x_n}{\frac{1}{n^\alpha}} = \frac{f\left(\frac{1}{n}\right)}{\left(\frac{1}{n}\right)^\alpha}$

Pp. că $(\exists) \lim\limits_{\substack{x \to 0 \\ x > 0}} \frac{f(x)}{x^\alpha} \overset{not}{=} L \in (0, +\infty)$

$a_n = \frac{1}{n} \to 0$, $\neq 0$ $\Rightarrow$

$\Rightarrow \frac{f\left(\frac{1}{n}\right)}{\left(\frac{1}{n}\right)^\alpha} \to L$

Ex Convergența seriei

$$\sum_{n \geq 1} \underbrace{n^a \left(\frac{1}{n} - \operatorname{arctg} \frac{1}{n}\right)}_{= x_n}, \quad a \in \mathbb{R}$$

$(\forall) x > 0 \Rightarrow \operatorname{arctg} x < x \Rightarrow x_n > 0$, $(\forall) n \geq 1$

$x_n = \frac{\frac{1}{n} - \operatorname{arctg} \frac{1}{n}}{\left(\frac{1}{n}\right)^a}$

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$$\frac{x_n}{\frac{1}{n^\alpha}} = \frac{\frac{1}{n} - \operatorname{arctg}\frac{1}{n}}{\left(\frac{1}{n}\right)^{\alpha+a}}$$

$1+x^2$

$$\lim_{\substack{x\to 0\\ x>0}} \frac{x - \operatorname{arctg} x}{x^{\alpha+a}} \overset{L'H}{=} \lim_{\substack{x\to 0\\ x>0}} \frac{1 - \frac{1}{1+x^2}}{(\alpha+a)\,x^{\alpha+a-1}} =$$

$\alpha + a > 0$

$$= \lim_{\substack{x\to 0\\ x>0}} \underbrace{\frac{1}{1+x^2}}_{\to 0} \cdot \frac{x^2}{(\alpha+a)\,x^{\alpha+a-1}} = 1 \cdot \lim_{\substack{x\to 0\\ x>0}} \frac{1}{(\alpha+a)\,x^{\alpha+a-3}} \Rightarrow$$

$\Rightarrow \alpha + a > 3$ Fie $\alpha = 3 - a$ $\Rightarrow$ ~~a > 3~~

$$\Rightarrow \lim_{\substack{x\to 0\\ x>0}} \frac{1}{(\alpha+a)\,x^{\alpha+a-3}} = \frac{1}{3}$$

$$\Rightarrow \lim_{\substack{x\to 0\\ x>0}} \frac{x - \operatorname{arctg} x}{x^{\alpha+a}} = \frac{1}{3} \Rightarrow \frac{\frac{1}{n} - \operatorname{arctg}\frac{1}{n}}{\left(\frac{1}{n}\right)^{\alpha+a}} \to \frac{1}{3}$$

$$\frac{x_n}{\frac{1}{n^\alpha}}$$

Pt. $\alpha = 3 - a \Rightarrow \frac{x_n}{\frac{1}{n^\alpha}} \to \frac{1}{3} \neq \infty \neq 0 \Rightarrow \sum_n x_n$ și $\sum_n \frac{1}{n^\alpha}$ au același nat.

$\alpha = 3 - a$

$\alpha > 1 \Leftrightarrow a < 2 \Rightarrow \sum_n x_n$ conv

$3 - a > 1$

$\alpha = 3 - a \leq 1 \Leftrightarrow a \geq 2 \Rightarrow \sum_n x_n$ div

(T) Criteriul integral

$f:[q,\infty)\to[0,\infty)$ descrescătoare, $q\in\mathbb{N}$

$\sum_{n\geq q} f(n)$ este conv. $\Leftrightarrow \underbrace{\int_q^n f(x)\,dx}_{a_n}$ conv.

(A) $n\geq q+1$

$a_{n+1}-a_n=\int_q^{n+1} f(x)-\int_q^n f(x)=\int_n^{n+1}\underbrace{f(x)}_{\geq 0}\geq 0$

$a_{n+1}\geq a_n$, (A) $n\geq q+1$

a_n este crescător

- a_n convergent $\Leftrightarrow$ a_n mărginit superior
- a_n divergent $\Leftrightarrow$ $a_n\to\infty$

$S_n=f(q)+f(q+1)+\dots+f(n)=\sum_{k=q}^{n} f(k)$

$n\geq q+1$

$a_n=\int_q^n f(x)\,dx=\int_q^{q+1} f(x)dx+\int_{q+1}^{q+2} f(x)dx\dots+\int_{n-1}^{n} f(x)dx$

$a_n=\sum_{k=q}^{n-1}\left(\int_k^{k+1} f(x)\,dx\right)$

$q\leq k\leq n-1$

$x\in[k,k+1]$, $k\leq x\leq k+1$

$f(k)\geq f(x)\geq f(k+1)$, (A) $x\in[k,k+1]$

$\int_k^{k+1}\underbrace{f(k)}_{f(k)}dx\geq\int_k^{k+1} f(x)dx\geq\int_k^{k+1}\underbrace{f(k+1)}_{f(k+1)}dx$

(A) k, $q\leq k\leq n-1$

$f(k+1)\leq\int_k^{k+1} f(x)\,dx\leq f(k)$

$\sum_{k=q}^{n-1} f(k+1)\leq\sum_{k=q}^{n-1}\int_k^{k+1} f(x)\,dx\leq\sum_{k=q}^{n-1} f(k)$

$$\sum_{m=q+1}^{n} f(k) \le a_n \le \sum_{k=q}^{n-1} f(k) \quad (\forall) n \ge q+1$$

$$\| \qquad\qquad \wedge$$

$$S_n - f(q) \qquad \sum_{k=q}^{n-1} f(k) + f(n) \overset{n^\circ}{=} S(n)$$

$S_n - f(q) \le a_n \le S_n$, $(\forall) n \ge q+1$

Pp: $\Leftarrow$ a_n convergent $\Rightarrow (\exists) M \in \mathbb{R}$ a.î. $a_n \le M$, $(\forall) n \ge q$

$S_n \le a_n + f(q) \le M + f(q)$ $(\forall) n \ge q+1 \Rightarrow S_n$ e mărginit superior

$\Rightarrow \sum_n f(n)$ conv

$\Leftarrow$ $\sum_n f(n)$ conv $\Rightarrow S_n$ mărginit sup.

$a_n \le S_n$, $(\forall) n \ge q+1 \Rightarrow a_n$ mărg superior

$(\exists) M$ a.î. $S_n \le M$

Prop $\sum_{n \ge 1} \frac{1}{n^\alpha}$, $\alpha \in \mathbb{R}$

1) Dacă $\alpha > 1 \Rightarrow \sum_{n \ge 1} \frac{1}{n^\alpha}$ conv

2) Dacă $\alpha \le 1 \Rightarrow \sum_{n \ge 1} \frac{1}{n^\alpha}$ div

Dem: $f(x) = \frac{1}{x^\alpha}$

Caz $\alpha \le 0$, $n^\alpha \le n^0 = 1$, $(\forall) n \ge 1$

$\frac{1}{n^\alpha} \ge 1$

Dacă prin absurd $\sum_n \frac{1}{n^\alpha}$ ar fi conv $\Rightarrow \lim_{n \to \infty} \frac{1}{n^\alpha} \to 0$

$\Rightarrow 0 \ge 1$ Fals

Caz: $\alpha > 0$

$f: [1, +\infty) \to (0, +\infty)$

$f(x) = \frac{1}{x^\alpha} = x^{-\alpha}, (\forall) x \geq 1$

$\Rightarrow$ e strict descrescătoare

$f'(x) = -\alpha x^{-\alpha-1} \Big/ < 0 \Rightarrow f$ descrescătoare
$\alpha > 0$

$\sum_{n \geq 1} \frac{1}{n^\alpha}$; $a_n = \int_1^n f(x)\,dx$

$$a_n = \int_1^n x^{-\alpha}\,dx = \begin{cases} \frac{x^{-\alpha+1}}{1-\alpha}\Big|_1^n, & \alpha \neq 1 \\ \ln x\Big|_1^n, & \alpha = 1 \end{cases}$$

$\alpha \neq 1 \Rightarrow a_n = \frac{1}{(1-\alpha) n^{\alpha-1}} - \frac{1}{1-\alpha} \to \begin{cases} \frac{1}{\alpha-1}, & \alpha > 1 \\ \infty, & 0 < \alpha < 1 \end{cases}$

~~$\alpha \leq 0$~~

~~$P: \alpha < 1 \quad n^{1-\alpha} > 1-\alpha$~~

~~$1-\alpha$~~

$\alpha = 1 \Rightarrow a_n = \ln n \to \infty$

Criteriul radicalului

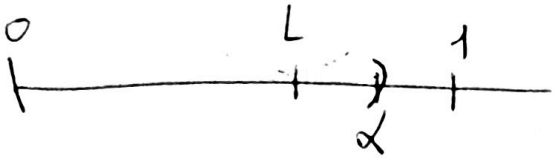
$x_n \geq 0, (\forall) n \geq p$

$\sqrt[n]{x_n} \to L \in [0, +\infty]$

1) $L < 1 \Rightarrow \sum_n x_n$ conv

2) $L > 1 \Rightarrow x_n \to \infty \Rightarrow \sum_n x_n$ div

1) $0 \le L < 1$



$(\exists)\ \alpha \in (L, 1)$, $\alpha = \frac{1+L}{2}$

$0 \le L < \alpha < 1$, $\alpha \in [0, 1)$

$(\exists)\ q \in \mathbb{N}$, $q \ge p$ a.î. $(\forall)\ n \ge q$, $\sqrt[n]{x_n} \in (-\infty, \alpha)$

$(\forall)\ n \ge q$, $\sqrt[n]{x_n} < \alpha$

$0 \le x_n \le \alpha^n$

$\alpha \in [0,1) \Rightarrow \sum_n \alpha^n$ conv. $\Big| \Rightarrow \sum_n x_n$ conv.

---

2) $L > 1$

Fie $\alpha \in (1, L)$, $\alpha = \frac{1+L}{2}$, dacă $L \ne \infty$

$L > \alpha > 1$

$\alpha = 2$, dacă $L = \infty$

$(\exists)\ q \ge p$ a.î. $(\forall)\ n \ge q \Rightarrow \sqrt[n]{x_n} \in (\alpha, \infty)$

$\sqrt[n]{x_n} > \alpha$, $\alpha > 1 \Rightarrow x_n > \alpha^n$, $(\forall)\ n \ge q \Big| \Rightarrow x_n \to \infty$ ($\alpha^n \to \infty$)

(T) Cri Roabe-Duhamel

$x_n > 0$, $(\forall)\ n \ge p$

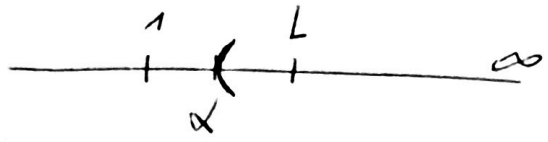
$n\left(\frac{x_n}{x_{n+1}} - 1\right) \longrightarrow L \in \overline{\mathbb{R}}$

1) Dacă $L > 1 \Rightarrow \sum_n x_n$ conv

2) Dacă $L < 1 \Rightarrow \sum_n x_n$ div

Dem:

1)



Fie $\alpha \in (1, L)$, $L > \alpha > 1$

$(\exists)\ q \geq p$ a.î. $n\left(\frac{x_n}{x_{n+1}} - 1\right) \in (\alpha, \infty)$, $(\forall)\ n \geq q+1$

$$\frac{n x_n - n x_{n+1}}{x_{n+1}} > \alpha \iff n x_n - n x_{n+1} > \alpha x_{n+1} \ \Big| - x_{n+1} \iff$$

$$\iff n x_n - (n+1) x_{n+1} > (\alpha - 1) x_{n+1}, \ (\forall)\ n \geq q$$

$p = q$: $q x_q - (q+1) x_{q+1} > (\alpha - 1) x_{q+1}$

$(q+1) x_{q+1} - (q+2) x_{q+2} > (\alpha - 1) x_{q+2}$

$\vdots$

$(n-1) x_{n-1} - n x_n > (\alpha - 1) x_n$

$n x_n - (n+1) x_{n+1} > (\alpha - 1) x_{n+1}$

$$q x_q - (n+1) x_{n+1} > \underbrace{(\alpha - 1)}_{>0} \underbrace{(x_{q+1} + \dots + x_n + x_{n+1})}_{>0}$$

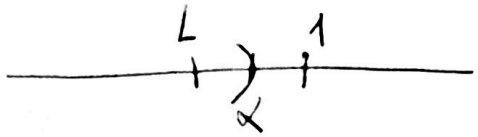
$$q x_q \geq q x_n - (n+1) x_{n+1} > (\alpha - 1)(x_{q+1} + \dots + x_n)$$

$$\underbrace{x_{q+1} + \dots + x_n}_{\text{șirul sumelor parțiale pt } \sum_{n \geq q+1} x_n} < \frac{q x_q}{\alpha - 1}, \ (\forall)\ n \geq q+1$$

, e mărginit superior

$\Rightarrow \sum_{n \geq q+1} x_n$ conv $\Rightarrow \sum_{n \geq 1} x_n$ conv.

2) $L < 1$



$-\infty \leq L$

$\alpha \in (L, 1)$

(3) $\exists q \geq p$ a.î. $(\forall) n \geq q$: $\dfrac{n x_n - n x_{n+1}}{x_{n+1}} \in (-\infty, \alpha)$ $\quad \alpha$

$\alpha < 1$

$n x_n - n x_{n+1} < \alpha x_{n+1} \mid - x_{n+1}$

$n x_n - (n+1) x_{n+1} < \underbrace{(\alpha - 1)}_{<0} x_{n+1} \mid \cdot \dfrac{1}{\alpha - 1}$

$x_{n+1} < \underbrace{\dfrac{1}{1-\alpha}}_{>0} \left((n+1) x_{n+1} - n x_n \right) \; | (\forall) n \geq q$

~~$x_{n+1} < \frac{1}{1-\alpha} ((n+1) x_{n+1} - n x_n)$~~

~~$x_{n+1} \geq q+1$~~

~~$x_n \leq \frac{1}{1-\alpha} (n x_n - (n-1) x_{n-1}) \; (\forall) n \geq q+1$~~

$(n+1) x_{n+1} - n x_n > 0, \; (\forall) n \geq q$

$(n+1) x_{n+1} > n x_n, \; (\forall) n \geq q$

Şirul $(n x_n)_{n \geq q}$ este strict crescător

$(\forall) n \geq q$: $n x_n \geq q x_q$

$(\forall) n \geq q$: $x_n \geq \underbrace{q x_q}_{>0} \cdot \dfrac{1}{n} > 0$

$\sum_n \frac{1}{n}$ este divergent $\Rightarrow$

$\Rightarrow \sum_n q x_q \cdot \frac{1}{n}$ div.

$\Rightarrow \Leftarrow$ $\sum_n x_n$ este divergent

~~Seria~~ Obs: $\frac{1}{n} \to 0$, $\sum_n \frac{1}{n}$ div.

Cri. necesar:

Dacă $\sum_n z_n$ e conv $\Rightarrow z_n \to 0$

Serii:

Def: $z_n \in \mathbb{C}$

$\sum_n z_n$ s.n. absolut convergentă dacă seria $\sum_n |z_n|$ este convergentă.

Def: $(z_n)_{n \geq p}$ șir de nr. complexe

șir Cauchy dacă $(\forall)\ \varepsilon > 0, (\exists)\ n_\varepsilon \geq p, n_\varepsilon \in \mathbb{N}$ a.î. $(\forall)\ n \geq n_\varepsilon$ și $(\forall)\ p \geq 1, p \in \mathbb{N}$ avem $|z_{n+p} - z_n| < \varepsilon$

(T) $(z_n)_{n \geq p}$ este șir Cauchy $\Leftrightarrow$ $(z_n)_{n \geq p}$ este convergent

(T) Criteriul Cauchy

$S_n = z_p + \dots + z_n$, (S_n) șir conv
$|S_{n+p} - S_n| < \varepsilon$
$S_{n+p} = z_p + \dots + z_n + z_{n+1} + \dots + z_{n+p}$

seria $\sum_{n \geq p} z_n$ este convergentă $\Leftrightarrow$ $(\forall)\ \varepsilon > 0, (\exists)\ n_\varepsilon \geq p, n_\varepsilon \in \mathbb{N}$ a.î. $(\forall)\ n \geq n_\varepsilon$ și $(\forall)\ p \geq 1, p \in \mathbb{N}$ avem: $|z_{n+1} + z_{n+2} + \dots + z_{n+p}| < \varepsilon \Rightarrow$

$\Rightarrow \forall \varepsilon \left| \sum_{k=n+1}^{n+p} z_k \right| < \varepsilon$

Prop: Dacă $\sum_n z_n$ este ~~conv~~ absolut conv, atunci $\sum_n z_n$ este și conv.

Dem: $\sum_{n\geq p} z_n$, $\sum_{n\geq p} |z_n|$ conv $\overset{\text{C. Cauchy}}{\Rightarrow}$ $(\forall)\,\varepsilon>0, (\exists)\,n_\varepsilon$,

a.î. $(\forall)\, n\geq n_\varepsilon, (\forall)\, p\geq 1 \Rightarrow \underbrace{|z_{n+1}|+|z_{n+2}|+\dots+|z_{n+p}|}_{\geq |z_{n+1}+z_{n+2}+\dots+z_{n+p}|} < \varepsilon$

$\rightarrow \sum_n z_n$ conv.

$z_n = x_n + i y_n$, $x_n \in \mathbb{R}$, $y_n \in \mathbb{R}$, $n\geq p$

$\sum_{n\geq p} z_n$

$z_p + z_{p+1} + \dots + z_n = \underbrace{(x_p + x_{p+1} + \dots + x_n)}_{\in \mathbb{R}} + i \underbrace{(y_p + \dots + y_n)}_{\in \mathbb{R}}$

$\sum_{n\geq p} z_n$ conv $\Leftrightarrow (z_p + \dots + z_n)_{n\geq p}$ conv $\Leftrightarrow$ $\begin{cases} x_p + \dots + x_n \text{ conv} \\ y_p + \dots + y_n \text{ conv} \end{cases}$ $\Leftarrow$

Prop: $z_n = x_n + i y_n$, $(\forall) n\geq p$, $y_n, x_n \in \mathbb{R}$, $(\forall) n\geq p$

Atunci: $\sum_n z_n$ este conv $\Leftrightarrow$ serile $\sum_n x_n$ și $\sum_n y_n$ sunt conv.

Ex: Studiați conv. seriei $\sum_{n\geq 1} \frac{1}{n+i}$

$z_n = \frac{1}{n+i} \overset{n-i}{=} \underbrace{\frac{n}{n^2+1}}_{x_n} + i \underbrace{\frac{-1}{n^2+1}}_{y_n}$

$x_n = \frac{n}{n^2(1+\frac{1}{n^2})} = \frac{1}{n(1+\frac{1}{n^2})} = \frac{1}{n}\cdot\frac{1}{1+\frac{1}{n^2}}$

$$\frac{x_n}{\frac{1}{n}} = \frac{1}{1+\frac{1}{n^2}} \longrightarrow 1 \neq 0 \Rightarrow \sum_n \frac{1}{n} \text{ div} \Rightarrow \sum_n x_n \text{ div} \Rightarrow$$

$$\Rightarrow \sum_n y_n \text{ div.}$$

Ex: $\sum_{n\geq 1} \underbrace{\frac{\sin\sqrt{n!+1}}{n\sqrt{n}}}_{=x_n}$

$$|x_n| = \frac{|\sin\sqrt{n!+1}|}{n\sqrt{n}} \leq \frac{1}{n^{\frac{3}{2}}}, (\forall) n \geq 1$$

$\frac{3}{2} > 1 \Rightarrow \sum_{n\geq 1} \frac{1}{n^{\frac{3}{2}}}$ e conv

$\Rightarrow \sum_n |x_n|$ convergentă $\Rightarrow$

$\Rightarrow \sum_n x_n$ este absolut conv. $\Rightarrow$

$\Rightarrow \sum_n x_n$ e conv.

(T) Cri. lui Dirichlet

$\sum_{n\geq q} a_n z_n$ $\quad a_n \in \mathbb{R}$, $z_n \in \mathbb{C}$

Pp că $(\exists)\ N \in \mathbb{N}, N \geq q$ a.î. $a_{n+1} \leq a_n\ (\forall) n \geq N$

Pp că $a_n \to 0$

Pp că $(\exists)\ M > 0$ a.î. $\left|\sum_{k=q}^{n} z_k\right| \leq M, (\forall) n \geq q$

$|z_q + z_{q+1} + \ldots + z_n|$

At $\Rightarrow \sum_n a_n z_n$ este conv

Obs $\lim_{n\to\infty} a_n = 0 \leq a_n, (\forall) n \geq N$

~~$\sup$~~ $\inf\{a_n \mid n \geq N\}$

(T) Crit. Leibniz

Fie $(a_n)_{n\geq 0}$ șir de nr. reale

Pp. că $(\exists)\ N \geq 0$, $N \in \mathbb{N}$, a.î. $(\forall)\ n \geq N$, avem $a_{n+1} \leq a_n$ și $a_n \to 0$.

Atunci $\sum_n (-1)^n a_n$ este conv. și $\sum_n (-1)^{n-1} a_n$ este conv.

Ex: Ex. de serie convergentă care nu este absolut conv.

$$\sum_{n\geq 1} \frac{(-1)^n}{n}$$

$n+1 > n$, $\frac{1}{n+1} < \frac{1}{n}$, $(\forall)\ n \geq 1$

Fie $a_n = \frac{1}{n} \to 0$ $\quad a_{n+1} < a_n$

Leibniz $\Rightarrow$ $\sum_n (-1)^n \cdot \frac{1}{n}$ conv

$\left|\frac{(-1)^n}{n}\right| = \frac{1}{n} \Rightarrow \sum_n \left|\frac{(-1)^n}{n}\right|$ este div $\Rightarrow$

$\Rightarrow \sum_n \frac{(-1)^n}{n}$ nu este absolut conv.

Ex: $\sum_{n\geq 1} (-1)^n \cdot \frac{n^2}{n^3 + 500.000}$ Studiați Conv

$$a_n = \frac{n^2}{n^3 + 500.000} \longrightarrow 0$$

$\sum_{n\geq 1} (-1)^n \cdot a_n$ Fie $f(x) = \frac{x^2}{x^3 + 5 \cdot 10^5}$, $x \geq 1$

$a_n = f(n)$, $(\forall)\ n \geq 1$ $\quad f: [1, +\infty) \to \mathbb{R}$

$$f'(x) = \frac{2x(x^3+5\cdot10^5) - x^2\cdot 3x^2}{(x^3+5\cdot10^5)^2}$$

$$= \frac{x(10^6-x^3)}{(x^3-5\cdot10^5)^2} \leq 0 \quad (\forall)\, x \geq 100$$

$10^6 \leq x^3 \Rightarrow x \geq 10^2 = 100 \Rightarrow x^3 \geq 10^6$

f este desc. pe $[100, \infty)$

$n \geq 100$, $n+1 > n \geq 100$

$f(n+1) \leq f(n)$

$\| \quad \|$

$a_{n+1} \quad a_n$

$(\forall)\, n \geq 100 \Rightarrow a_{n+1} \leq a_n \mid \Rightarrow a_n \to 0 \underset{\text{Conform } n}{\underset{\text{C. Leibniz}}{\Longrightarrow}} \sum (-1)^n a_n$ conv

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(T) Cr. Abel

$\sum_{n \geq q} a_n \cdot z_n$, $a_n \in \mathbb{R}$, $(\forall)\, n \geq q$

$z_n \in \mathbb{C}$, $(\forall)\, n \geq q$.

Pp că a_n este conv. și că $(\exists)\, N \geq q$ a.î. $a_{n+1} \leq a_n$, $(\forall)\, n \geq N$

(sau $a_{n+1} \geq a_n$, $(\forall)\, n \geq N$)

Pp că $\sum_n z_n$ este conv

Atunci $\sum_n a_n z_n$ este conv

Dem. temă. ooo

Se aplică C. lui Dirichlet:

a_n conv $\Rightarrow (\exists) \lim_{n\to\infty} a_n = L \in \mathbb{R}$

A_n descr și $A_n \to 0$

Dirichlet:

$A_n = a_n - L$ (dacă $a_n \searrow$)

($A_n = L - a_n$ (dacă $a_n \nearrow$))

Ex: Studiați conv. seriei:

$$\sum_{n\geq 1} \frac{\cos(nx)}{\sqrt{n}}, \quad x \in \mathbb{R}, \; x \notin \{2k\pi \mid k \in \mathbb{Z}\}$$

Dirichlet: $a_n = \frac{1}{\sqrt{n}} \to 0$

$\sqrt{n+1} > \sqrt{n}$, $\frac{1}{\sqrt{n+1}} < \frac{1}{\sqrt{n}}$

$a_{n+1} < a_n, (\forall) n \geq 1$

$z_n = \cos(nx)$

$S_n = (z_1 + z_2 + \dots + z_n) = \cos x + \cos 2x + \dots + \cos nx \in \mathbb{R}$

Fie $T_n = \sin x + \sin 2x + \dots + \sin nx \in \mathbb{R}$

$S_n + i T_n = (\cos x + i \sin x) + (\cos 2x + i \sin 2x) + \dots + (\cos nx + i \sin nx)$

$(\cos x + i \sin x)^k = \cos kx + i \sin kx$

$v = \cos x + i \sin x$

$|v| = \sqrt{\cos^2 x + \sin^2 x} = 1$

$S_n + i T_n = v^1 + v^2 + \dots + v^n$

$v - 1 = \sin x + \cos x \neq 1$

(Pp. că $v = 1 \Rightarrow \cos x + i \sin x = 1 + i \cdot 0 \Leftrightarrow \cos x = 1 \\ \sin x = 0$ $= \cos 0 \Leftrightarrow x \in \{2k\pi \mid k \in \mathbb{Z}\}$

$\Uparrow$

$x = 2k\pi, k \in \mathbb{Z}$

$\cos x + i \sin x = 1 \Leftrightarrow x \in \{2k\pi \mid k \in \mathbb{Z}\}$)

$S_n + i T_n = \frac{v(v^n - 1)}{v - 1}, |v| = 1$

$|S_n + i T_n| = \frac{\overbrace{|v|}^{=1} \cdot \overbrace{|v^n - 1|}^{\leq 2}}{|v - 1|} \leq \frac{1 \cdot 2}{|v-1|}, (\forall) n \geq 1$

$|v^n + (-1)| \leq |v^n| + |-1| = |v|^n + 1 = 2$

$(\forall) n \geq 1$ avem $|S_n + iT_n| \leq \frac{2}{|\varepsilon - 1|}$

$|S_n| \leq \sqrt{S_n^2 + T_n^2}$

$|T_n| \leq \frac{2}{|\varepsilon - 1|}$, $(\forall) n \geq 1$

$|S_n| \leq \frac{2}{|\varepsilon - 1|}$, $(\forall) n \geq 1$

$|\cos x + \cos 2x + \ldots + \cos nx| = Z_n$

$a_n = \frac{1}{\sqrt{n}} \to 0$, $a_{n+1} \leq a_n$ $(\forall) n \geq 1$

Dirichlet

$\Rightarrow \sum_n a_n Z_n$ conv. $\Rightarrow \sum_n \frac{1}{\sqrt{n}} \cos nx$ convergent

$\sum_{k=0}^{n} a_n x^n = a_0 + a_1 x + \dots + a_n x^n$

Șiruri de funcții

Def: A - mulțime oarecare $\neq \emptyset$

$\forall n \in \mathbb{N}$. Fie: $f_n : A \to \mathbb{C}$ ($\mathbb{R}$), $n \geq \ell$

1) Șirul de funcții $(f_n)_{n \geq \ell}$ s.n. <u>simplu convergent</u> (punctual conv.) (S.C.) (P.C.) la f dacă $(\forall) x \in A$, șirul $(f_n(x))_{n \geq \ell}$ converge la $f(x)$.

<u>Notăm</u>: $f_n \xrightarrow{\text{simplu}} f$ sau $f_n \xrightarrow{S.C.} f$ sau $f_n \xrightarrow{P.C.} f$

2) Șirul $(f_n)_{n \geq \ell}$ s.n. <u>uniform conv. la f</u> dacă (U.C.)
$(\forall) \varepsilon > 0$, $(\exists) n_\varepsilon \geq \ell$ ($n_\varepsilon \in \mathbb{N}$) a.î. $(\forall) n \geq n_\varepsilon$ și $(\forall) x \in A$, avem $|f_n(x) - f(x)| < \varepsilon$
(n_ε nu depinde de x)

<u>Notații</u> $f_n \xrightarrow{\text{uniform}} f$ sau $f_n \xrightarrow{U.C.} f$.

<u>Obs</u>: Dacă $f_n \xrightarrow{U.C.} f \Rightarrow f_n \xrightarrow{S.C.} f$.

<u>Prop:</u> $f_n : A \to \mathbb{C}$, $f : A \to \mathbb{C}$

<u>At:</u>

a) $f_n \xrightarrow[A]{U.C.} f \iff |f_n - f| \xrightarrow[A]{U.C.} 0$

b) $f_n \xrightarrow[A]{U.C.} 0 \iff |f_n| \xrightarrow[A]{U.C.} 0$

c) Dacă $f_n \xrightarrow[A]{U.C.} f \Rightarrow |f_n| \xrightarrow[A]{U.C.} |f|$

$$\big||f_n(x)| - |f(x)|\big| \leq |f_n(x) - f(x)|$$

Prop: $f_n, f: A \to \mathbb{C}$, $n \geq 0$

Pp. că $f_n \xrightarrow[A]{U.C.} f$.

Atunci $(\forall)(f_{k_n})_{n\geq 1}$ un subșir al șirului de funcții $(f_n)_n$ avem $f_{k_n} \xrightarrow[A.]{U.C.} f$.

Prop: Fie $f_n, f: A \to \mathbb{C}$. Atunci $f_n \xrightarrow[A]{U.C.} f \iff$

$\iff \sup_{x\in A} |f_n(x) - f(x)| \xrightarrow{n\to\infty} 0$

OBS: $f_n \xrightarrow[A]{U.C.} f \Rightarrow f_n \xrightarrow[A]{S.C.} f$.

OBS: Fie $f_n: A \to \mathbb{R}$, $n \geq 0$

Dacă $(\exists) x \in A$ a.î. ~~$\lim_{n\to\infty}$~~ $f_n(x)$ nu e convergent at. f_n nu C.S. pe $A \Rightarrow f_n$ nu C.U. pe A.

Pp. că $(\forall) x \in A$ $(\exists) \lim_{n\to\infty} f(x) \overset{not}{=} f(x) \in \mathbb{C} \Rightarrow$

$\Rightarrow f_n \xrightarrow[A]{C.S.} f$

Se dă: $\sup_{x\in A} |f_n(x) - f(x)| \to 0$ atunci $f_n \xrightarrow[A]{U.C.} f$.

1) Dacă

2) Dacă $\sup_{x\in A} |f_n(x) - f(x)| \not\to 0$ atunci $f_n \xrightarrow[A]{U.C.} f$.

- Dacă $\sup_{x\in A} |f_n(x) - f(x)| \longrightarrow L \in \bar{\mathbb{R}}$ și $L \neq 0 \Rightarrow f_n \xrightarrow[pe\ A]{U.C.} f$.
- Dacă $(\not\exists) \lim_{n\to\infty} \sup_{x\in A} |f_n(x) - f(x)| \Rightarrow f_n \xrightarrow[A]{C.U.} f$.
- Dacă $(\exists) k_n$ un șir strict crescător de numere nat ≥ 0 a.î. ~~$\lim_{n}$~~ $\sup_{x\in A} |f_{k_n}(x) - f(x)| = \infty$ atunci $f_n \xrightarrow[A]{C.U.} f$.

Prop: Fie $f_n, f: A \to \mathbb{C}$, $n \geq 0$

Pp că $(\exists)$ un șir $(a_n)_{n \geq N}$, $N \geq 0$, a.î. $(\forall) n \geq N$, $(\forall) x \in A$ avem: $|f_n(x) - f(x)| \leq a_n$

Dacă $a_n \to 0 \Rightarrow f_n \xrightarrow[A]{u.c.} f$.

$0 \leq \sup_{x \in A} |f_n(x) - f(x)| \leq a_n \downarrow 0$, $(\forall) n \geq N$

Prop: Pp. că $f_n \xrightarrow[A]{u.c.} f$. Atunci $(\forall)$ un șir de elemente din A, $(x_n)_{n \geq N}$ avem $[f_n(x_n) - f(x_n)] \to 0$

sau

$|f_n(x_n) - f(x_n)| \to 0$

Corolar: $f_n \xrightarrow[A]{s.c.} f$

Dacă $(\exists)$ $x_n \in A$ un șir a.î. $|f_n(x_n) - f(x_n)| \not\to 0$

atunci $f_n \xrightarrow[A]{u.c.} \not f$

Ex: $f_n(x) = \frac{nx^2}{x^3 + n^3}$ $(n \geq 1)$

Studiu. c. și s.c.

(a) pe $[0,4]$ (b) pe $[4, \infty)$

$\lim_{n \to \infty} \frac{nx^2}{n^3 + x^3} = 0$ $(\forall) x \geq 0$ $\Rightarrow f_n(x) \xrightarrow[[0,4]]{\text{simplu}} f$; $f_n(x) \xrightarrow[[4,+\infty)]{c.s.} f$ $\Big/ \Rightarrow$

$\Rightarrow f_n \xrightarrow[[0,+\infty)]{s.c.} f$

$g_n(x) = |f_n(x) - \underset{0}{f(x)}| = \frac{n x^2}{x^3 + n^3}$, $(\forall)\, n \geq 1$, $x \geq 0$

$g_n'(x) = n \frac{2x(x^3+n^3) - x^2 \cdot 3x^2}{(x^3+n^3)^2} = \frac{x(2n^3 - x^3)}{(x^3+n^3)^2} \cdot n$

$g_n'(x) = \frac{nx(2n^3 - x^3)}{(x^3+n^3)^2}$

$x = 0$

$x^3 = 2n^3$

$x = n\sqrt[3]{2}$

$n\sqrt[3]{2} > 4$

$2n^3 > 16 \cdot 4$

$n^3 > 32$

$4^3 = 64 > 32$ $\quad (\forall)\, n \geq 4 : n\sqrt[3]{2} \in (4, +\infty)$

(4) $x \in [0,4]$, $n \geq 4$

x	0		4		$n\sqrt[3]{2}$
$g_n'(x)$	0	$+$	$+$	$+$	0
$g_n(x)$	$g_n(0)$	$\nearrow$	$g_n(4)$		

Cum $x < n\sqrt[3]{2} \Rightarrow x^3 < 2n^3$

$\sup_{x \in [0,4]} g_n(x) = g_n(4) = \frac{n \cdot 4^2}{n^3 + 4^3} \to 0 \Rightarrow f_n \underset{[0,4]}{\overset{u.c.}{\to}} f = 0$

$g_n(x) = |f_n(x) - \underset{0}{f(x)}| \leq \frac{n \cdot 4^2}{n^3 + 4^3}$

b) $x \in [4, \infty)$, $n \geq 4$

x	4		$n\sqrt[3]{2}$		∞
$f_n'(x)$	+	+ +	0	- - -	
$f_n(x)$		↗	$f_n(n\sqrt[3]{2})$	↘	

$$\sup_{x \in [4,\infty)} f_n(x) = f_n(n\sqrt[3]{2}) = \frac{n \cdot \sqrt[3]{4}\, n^3}{2n^3 + n^3} = \frac{\sqrt[3]{4}}{3} \to \frac{\sqrt[3]{4}}{3} \neq 0$$

$$\Rightarrow f_n \xrightarrow[[4,\infty)]{u.c} f$$

Atlel:

$$f_n \xrightarrow[[0,4]]{s.c} 0 = f \quad ; \quad f_n \xrightarrow[[4,\infty)]{s.c} 0 = f$$

a) $x \in [0,4]$, $0 \leq x \leq 4$

$$|f_n(x) - f(x)| = \frac{n x^2}{x^3 + n^3} \leq \frac{n 4^2}{x^3 + n^3} \leq \frac{n 4^2}{n^3} = \frac{16}{n^2}$$

$$x^3 + n^3 \geq n^3$$

$(\forall)\ x \in [0,4]$, $(\forall)\ n \geq 1 \Rightarrow |f_n(x) - f(x)| \leq \frac{16}{n^2} \to 0$

$$\Rightarrow f_n \xrightarrow[[0,4]]{u.c} f = 0$$

b) $x \in [4,\infty)$, $x \geq 4$ $\quad f_n \xrightarrow[[4,+\infty)]{s.c} 0 = f$

$$|f_n(x) - f(x)| = \frac{n x^2}{x^3 + n^3}$$

$$|f_n(4) - f(4)| = \frac{n^3}{2n^3} = \frac{1}{2} \to \frac{1}{2} \neq 0 \Rightarrow f_n \xrightarrow[[4,\infty)]{c.u} f = 0$$

$x_n = n$, $n \geq 4$, $x_n \in [4, +\infty)$

Prop: i) Dacă $f_n \xrightarrow[A]{C.U} f$, $\alpha \in \mathbb{C}$ =>

=> $\alpha f_n \xrightarrow[A.]{U.C.} \alpha f$

ii) $f_n \xrightarrow[A]{U.C.} f$, $g_n \xrightarrow[A]{U.C.} g$, $\alpha, \beta \in \mathbb{C}$ =>

=> $(\alpha f_n + \beta g_n) \xrightarrow[A]{U.C.} (\alpha f + \beta g)$

iii) Fie $(\alpha_n)_n$ șir conv. la α de nr. complexe

$\alpha_n \xrightarrow{conv} \alpha$

Fie $f_n : A \to \mathbb{C}$, $n \geq p$

$f_n(x) = \alpha_n$, $(\forall)\ x \in A$ =>

=> atunci $f_n \xrightarrow[A]{U.C.} \alpha$

iv) Fie $\varphi : A \to \mathbb{C}$ o funcție

Fie $f_n : A \to \mathbb{C}$, $n \geq 0$

$f_n(x) = \varphi(x)$, $(\forall)\ x \in A$, $(\forall)\ n \geq 0$ => $f_n \xrightarrow[A.]{U.C.} \varphi$

Ex: $f_n, g_n : (0,1) \to \mathbb{R}$, $n \geq 1$

$f_n(x) = \frac{1}{n}$, $(\forall)\ x \in (0,1)$

$g_n(x) = \frac{1}{x}$, $(\forall)\ x \in (0,1)$

Avem: $f_n \xrightarrow[(0,1)]{U.C.} 0$

$g_n \xrightarrow[(0,1)]{U.C.} \frac{1}{x}$

$h_n(x) = \frac{1}{nx} \xrightarrow[(0,1)]{S.C.} 0 = h(x)$

Fie $h_n : (0,1) \to \mathbb{R}$, ~~$h_n(x) =$~~

$h_n(x) = f_n(x) \cdot g_n(x) = \frac{1}{nx}$, $(\forall)\ x \in (0,1)$

$|h_n(x) - h(x)| \geq \frac{1}{nx}$; $x_n = \frac{1}{n} \in (0,1)$

$\|h_n - h\| = 0$

$|h_n(\frac{1}{n}) - h(\frac{1}{n})| = 1 \to 1 \neq 0$ =>

=> $h_n \xrightarrow[(0,1)]{C.U} h$

(T) Transfer de continuitate

Fie $f_n : Y \subset \mathbb{R} \to \mathbb{C}$ funcții continue (pe Y), $(\forall) n \geq 0$

Pp. că $f_n \xrightarrow[Y]{u.c.} f$, $f : Y \to \mathbb{C}$

Atunci f este și ea continuă (pe Y)

$f(x) = \lim_{n\to\infty} f_n(x)$, $a \in Y$.

f continuă pe $a \Rightarrow$

$\Rightarrow f(a) = \lim_{x\to a} f(x)$

$\lim_{n\to\infty} f_n(a) = \lim_{x\to a} \left(\lim_{n\to\infty} f_n(x) \right)$

$\lim_{n\to\infty} \left(\lim_{x\to a} f_n(x) \right)$

(T) Fie $f_n : [a,b] \to \mathbb{C}$ funcție continue, $(\forall) n \geq 0$

Pp. că $f_n \xrightarrow[[a,b]]{u.c.} f$, $f : [a,b] \to \mathbb{C}$

(deci f este continuă)

Atunci $\left(\int_a^b f_n(x)\,dx \right)_{n\geq 0}$ este convergent și

$\lim_{n\to\infty} \int_a^b f_n(x)\,dx = \int_a^b f(x)\,dx$

$\int_a^b \left(\lim_{n\to\infty} f_n(x) \right) dx$

$\int_a^b f_n(x)\,dx \longrightarrow \int_a^b f(x)\,dx$

(T) $f_n: J \to \mathbb{C}$, funcții derivabile

Pp. că $f_n \xrightarrow[J]{c.s.} f$ și $f_n' \xrightarrow[J]{c.u.} g$ și $f, g: J \to \mathbb{C}$

- Atunci f este derivabilă și $f'(x) = g(x)$ $(\forall)\ x \in J$
- Dacă J este mărginit atunci $f_n \xrightarrow[J]{u.c.} f$

$$f'(x) = g(x)$$

$$\left(\lim_{n\to\infty} f_n\right)' = \lim_{n\to\infty} (f_n')$$

Ex: $f_n: [0,1] \to \mathbb{R}$, $n \geq 1$

$$f_n(x) = \frac{n^2 + \sin x}{x^{100} + n^2x^2 + ne^x + n^2}, \quad (\forall)\ x \in [0,1]$$

a) Este f_n uniform conv?

b) Calc. $\lim_{n\to\infty}\left(\int_0^1 \frac{n^2+\sin x}{x^{100}+n^2x^2+ne^x+n^2}\,dx\right)$

a) $f_n(x)$ sunt continue pe $[0,1]$, $(\forall)\ n \geq 1$

$$f_n(x) = \frac{n^2+\sin x}{n^2(x^2+1)+ne^x+x^{100}} \xrightarrow{n\to\infty} \underbrace{\frac{1}{x^2+1}}_{=f(x)} \quad (\forall)\ x \in [0,1]$$

$f_n \xrightarrow[[0,1]]{s.c.} f$ $\qquad |f_n(x) - f(x)| = \left|\frac{n^2+\sin x}{n^2(x^2+1)+ne^x+x^{100}} - \frac{1}{x^2+1}\right| =$

$$= \frac{|(\sin x)(x^2+1) - ne^x - x^{100}|}{(n^2(x^2+1)+ne^x+x^{100})(x^2+1)} \leq \frac{|\sin x|(x^2+1) + |-ne^x| + x^{100}}{(x^2+1)(\underbrace{n^2(x^2+1)}_{\geq 0} + \underbrace{ne^x}_{\geq 0} + \underbrace{x^{100}}_{\geq 0})} \leq$$

$$\leq \frac{\overbrace{|\sin x|}^{\leq 1}\,(x^2+1) + n\,\overbrace{(e^x + x^{100})}^{\leq 2}}{(x^2+1)\left(n^2 (x^2+1)\right)} \leq 1$$

$$\leq \frac{1\cdot(1+1) + n\cdot e + 1}{(x^2+1)^2 \cdot n^2}, \quad x \geq 0 \Rightarrow x^2+1 \geq 1 \Rightarrow \frac{1}{(x^2+1)^2} \leq 1$$

$$\Rightarrow \frac{2 + ne + 1}{(x^2+1)^2 \cdot n^2} \leq 1 \cdot \frac{3+en}{n^2}$$

$$|f_n(x) - f(x)| \leq \frac{en+3}{n^2} \longrightarrow 0 \quad (\forall)\, x \in [0,1],\ (\forall)\, n \geq 1$$

$$\Rightarrow f_n \xrightarrow[[0,1]]{u.c.} f.$$

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b) $$\lim_{n\to\infty} \int_0^1 f_n(x)\,dx = \int_0^1 f(x)\,dx = \int_0^1 \frac{1}{x^2+1}\,dx = \operatorname{arctg}\frac{\pi}{4}$$

Ex: $f_n : [0,1] \longrightarrow \mathbb{R}$ $\quad f_n(x) = n x e^{-nx^2}$, $(\forall)\, x \in [0,1]$

(a) Arătați că f_n este S.C. la o funcție continuă

(b) Arătați că: $(\exists)\ \lim_{n\to\infty}\left(\int_0^1 f_n(x)\,dx\right) \neq \int_0^1 \left[\lim_{n\to\infty}(f_n(x)\,dx)\right]$

(c) Este f_n unform conv. ???

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a) $f_n(0) = 0 \to 0$

$0 < x \leq 1$, $f_n(x) = n x e^{-nx^2} = \frac{nx}{e^{nx^2}} > 0$

$$\frac{f_{(n+1)}(x)}{f_n(x)} = \frac{(n+1)\,x\,e^{-x^2(n+1)}}{n x e^{-x^2 n}} = \frac{n+1}{n}\,e^{-x^2} \longrightarrow e^{-x^2} < 1$$

$\Rightarrow \sum_n f_n(x)$ convergent $\Rightarrow f_n(x) \longrightarrow 0$

$(\forall)\ x \in [0,1] \Rightarrow f_n(x) \longrightarrow 0 = f(x)$

$f: [0,1] \to \mathbb{R}$ → continuă

$\int_0^1 \left(\underbrace{\lim_{n\to\infty} f_n(x)}_{=0}\right) dx = 0$

$\int_0^1 f_n(x)\,dx = \int_0^1 \left((2n x) e^{-n x^2} \cdot -\frac{1}{2}\right) dx = -\frac{1}{2} e^{-n x^2} \Big|_0^1 =$

$= -\frac{1}{2} \cdot \frac{1}{e^n} + \frac{1}{2} \longrightarrow \frac{1}{2}$

⑨ $f_n(x) \xrightarrow[[0,1]]{\text{U.C.}} f(x)$ deoarece $\int_0^1 \lim_{n\to\infty} f_n(x)\,dx \neq \lim \int_0^1 \lim_{n\to\infty} f_n(x)$

$f_n: [0,1] \to \mathbb{R},\ f_n(x) = n(1-x)x^n = n x^n - n x^{n+1}$

a) Arătați că f_n converge simplu la o funcție continuă

b) $(\exists) \lim_{n\to\infty} \int_0^1 f(x)\,dx = \int_0^1 \lim_{n\to\infty} f_n(x)\,dx$

c) f_n nu e uniform. conv.

⓪ $f_n(0) = 0 \to 0$
$f_n(1) = 0 \to 0$

$\frac{n}{\left(\frac{1}{x}\right)^n}$, $n x^n \to 0$, $x \in (0,1)$ ⇙ $\frac{1}{x} > 1$

$(\forall)\ a > 1 \Rightarrow \frac{n}{a^n} \to 0$

$\lim_{y\to\infty} \frac{y}{a^y} \overset{L'H.}{=} \lim_{y\to\infty} \frac{1}{a^y \ln a} = \frac{1}{\infty} = 0 \Rightarrow \lim_{n\to\infty} \frac{n}{a^n} = 0$

$f_n(x) = (1-x) \cdot \frac{n}{\left(\frac{1}{x}\right)^n} \longrightarrow 0$ $\quad 0<x<1 \Rightarrow \frac{1}{x} > 1$

$\left(\frac{1}{x}\right)^n \to 0$

$(\forall)\ x \in [0,1] \Rightarrow f_n(x) \to 0 \Rightarrow f_n \xrightarrow[[0,1]]{S.C.} 0 = f$ e continuă

b) $\int_0^1 \left(\lim_{n\to\infty} f_n(x)\right) dx = 0$

"0

$\int_0^1 f_n(x)\,dx = \int_0^1 (nx^n - nx^{n+1})\,dx = \left(n\frac{x^{n+1}}{n+1} - \frac{n x^{n+2}}{n+2}\right)\Big|_0^1 =$

$= \frac{n}{n+1} - \frac{n}{n+2} \to 1 - 1 = 0 \Rightarrow$ [illegible]

$\Rightarrow \lim_{n\to\infty} \int_0^1 f_n(x)\,dx = \int_0^1 \lim_{n\to\infty} f_n(x)\,dx$

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c) $f_n(x) = n(1-x)x^n \to f(x) = 0$

$g_n(x) = |f_n(x) - f(x)| = \frac{nx^n - nx^{n+1}}{n(1-x)x^n}$

$x \in [0,1],\ n \geq 1$

$g_n'(x) = n(nx^{n-1} - (n+1)x^n) = nx^{n-1}(n+1)\left(\frac{n}{n+1} - x\right)$

$n \geq 2$ $\quad x = 0,\ 0 < x = \frac{n}{n+1} < 1$ / deci e altă cazuri de observat

x	0		$\frac{n}{n+1}$		1
$g_n'(x)$	0	$+ + +$	0	$- - -$	
$g_n(x)$		$\nearrow$	$g_n\left(\frac{n}{n+1}\right)$	$\searrow$	

$$\sup_{x\in[0,1]} g_n(x) = g_n\left(\frac{n}{n+1}\right)$$

$$g_n = |f_n(x) - f(x)|$$

$$\sup_{x\in[0,1]} g_n(x) = n\left(1 - \frac{n}{n+1}\right)\left(\frac{n}{n+1}\right)^n = \frac{n}{n+1}\cdot\frac{1}{\left(1+\frac{1}{n}\right)^n} \longrightarrow \frac{1}{e}\cdot 1 \neq$$

$$\Rightarrow f_n \xrightarrow[[0,1]]{\text{u.c.}} f$$

Serii de funcții

$\sum_{n\geq 0} f_n$ - serie de funcții, $f_n: A \neq \emptyset \to \mathbb{C}_{(R)}$, $n \geq 0$

$(f_n)_{n\geq 0}$ - șir de funcții

$S_n: A \to \mathbb{C}_{(R)}$, $S_n(x) = f_0(x) + f_{1}(x) + \dots + f_n(x)$,

$(\forall) x \in A$, $n \geq 0$

Def:

- Seria de funcții se numește simplu convergentă pe A dacă $(\forall) x \in A$, $\sum_{n\geq 0} f_n(x)$ este convergentă. (adică, $(S_n)_n$ converge simplu (pe A))

- Seria de funcții $\sum_{n\geq 0} f_n$ s.n. uniform convergentă (pe A) dacă șirul de funcț. $(S_n)_n$ este uniform convergent (pe A)

Proprietăți:

i) Dacă $\sum_{n\geq 0} f_n$ U.C. pe A și $(\forall) \alpha \in \mathbb{C} \Rightarrow$ $\sum_{n\geq 0} \alpha \cdot f_n$ e U.C. pe A

ii) Fie $\alpha, \beta \in \mathbb{C}$. Dacă $\sum_{n\geq 0} f_n$ și $\sum_{n\geq 0} g_n$ sunt U.C. pe A, atunci $\sum_{n\geq 0} (\alpha f_n + \beta g_n)$ e U.C.

$r, n, q \in \mathbb{N}$, $n > q$. $\sum_{n \geq q} f_n$, $f_n : A \to \mathbb{C}$

$S_n(x) = \underbrace{f_q(x) + \dots + f_{r-1}(x)}_{F(x)} + \underbrace{f_r(x) + \dots + f_n(x)}_{T_n(x)}$

$S_n(x) = F(x) + T_n(x)$, $(\forall)\ n \geq r$, $x \in A$ $\quad x \in A,\ n \geq r.$

$\sum_{n \geq q} f_n$ (1) $\qquad \sum_{n \geq r} f_n(x)$ (2)

Dacă (2) e U.C. pe $A \iff T_n$ e U.C. pe A.
$F(x)$ e. U.C. pe A (ca $F(x)$) $\Rightarrow$

$\Rightarrow F(x) + T_n(x) = S_n(x)$ este U.C. $\Rightarrow \sum_{n \geq q} f_n$ e. U.C.

Obs: $r, q \in \mathbb{N}$, $r > q$, $f_n : A \to \mathbb{C}$

Af.: $\sum_{n \geq q} f_n$ e. U.C. pe $A \iff \sum_{n \geq r} f_n$. este. U.C. pe A.

Vom nota $\sum_n f_n$ e. U.C. pe A
(sau mai e U.C. pe A)

(T) Fie $\sum_{n \geq q} f_n$ e U.C. pe $A \Rightarrow f_n \xrightarrow[\text{pe } A]{u.} 0$

Dem:

$S_n(x) = f_q(x) + \dots + f_{n-1}(x) + f_n(x)$ converge uniform pe $A \Rightarrow$

$\Rightarrow (\exists)\ S(x)$ a.î. $S_n \xrightarrow[\text{pe } A]{u.} S \Rightarrow S_{n-1} \xrightarrow[\text{pe } A]{u.} S \ / \Rightarrow$

$\Rightarrow \underbrace{S_n - S_{n-1}}_{f_n} \xrightarrow[\text{pe } A]{u.} S - S$

T. Transfer de continuitate pt. serii de funcții.

Fie $f_n: I \to \mathbb{C}$, f_n continue pe I, $(\forall) n \geq 0$

Dacă $\sum_{n \geq 0} f_n$ e U.C. pe I, at. $\underbrace{\sum_{n=0}^{\infty} f_n}_{\text{Sumă}}$ este cont. pe I.

Dem: $S_n: I \to \mathbb{C}$

$S_n(x) = \underbrace{f_0(x) + \dots + f_n(x)}_{\text{Continue pe } I}$, $(\forall) x \in I$, $n \geq 0$

$\Rightarrow S_n$ e cont. pe I, $(\forall) n \geq 0$.

$\sum_n f_n$ U.C pe $I \Rightarrow S_n$ e U.C. pe $I \Rightarrow$

$\Rightarrow \boxed{\lim_{n \to \infty} S_n}$ e cont. pe I

$\therefore \sum_{n=0}^{\infty} f_n$

Teremă: Integrare termen cu termen.

Fie $f_n: [a, b] \to \mathbb{C}$, f_n cont pe $[a, b]$, $(\forall) n \geq 0$

Pp. că: $\sum_{n \geq 0} f_n$ este U. C. pe $[a, b]$

At. seria $\sum_{n \geq 0} \left(\int_a^b f_n(x)\, dx \right)$ este conv. și

$$\int_a^b \left(\sum_{n=0}^{\infty} f_n(x) \right) dx = \sum_{n=0}^{\infty} \int_a^b f_n(x)\, dx$$

Dem: $S_n(x) = f_0(x) + \dots + f_n(x) \xrightarrow[p.p. [a,b]]{u} S(x)$

$S(x) = \sum_{n=0}^{\infty} f_n(x)$

Cum funcțiile sunt continue $\Rightarrow S_n : [a,b] \to \mathbb{R}$ e cont $\Rightarrow$

$\Rightarrow S_n \xrightarrow[[a,b]]{u} S$

Af. $\int_a^b S_n(x)\,dx$ este conv. și $\lim_{n\to\infty} \int_a^b S_n(x)\,dx = \int_a^b S(x)\,dx$

$\int_a^b S_n(x)\,dx = \int_a^b f_0(x)\,dx + \dots + \int_a^b f_n(x)\,dx$

Conv.

este șirul sumelor parțiale

p.t. seria $\sum_{n\geq 0} \left(\int_a^b f_n(x)\,dx\right)$

$\Rightarrow \sum_{n\geq 0} \left(\int_a^b f_n(x)\right)dx$ este convergentă.

$\lim_{n\to\infty} \int_a^b S_n(x)\,dx = \int_a^b S(x)\,dx$

$\lim_{n\to\infty} \left(\int_a^b f_0(x) + \dots + \int_a^b f_n(x)\,dx\right) \overset{\text{suma}}{=\!=} \sum_{n=0}^{\infty} \left(\int_a^b f_n x\,dx\right)$

$\int_a^b S(x)\,dx = \int_a^b \lim_{n\to\infty} \left(f_0(x) + \dots + f_n(x)\right)dx = \int_a^b \left(\sum_{n=0}^{\infty} f_n(x)\right)dx$

Teor: Derivare termen cu termen

Fie $f_n : J \to \mathbb{C}$ funcții derivabile pe J, $(\forall) n \geq 0$

Pp că:

$\sum_{n \geq 0} f_n$ converge simplu pe J și $\sum_{n \geq 0} f_n'$ converge uniform pe J

Atunci: $\sum_{n=0}^{\infty} f_n$ este derivabilă pe J și $\left(\sum_{n=0}^{\infty} f_n\right)' = \sum_{n=0}^{\infty} f_n'$ pe J.

(suma seriei)

(+) Dacă J este mărginit $\Rightarrow \sum_{n \geq 0} f_n$ e U.C. pe J.

$S_n(x) = \underbrace{f_0(x) + \dots + f_n(x)}_{\text{derivabilă}}$, $x \in J$, $n \geq 0$

$\Rightarrow S_n$ e derivabilă

$S_n'(x) = f_0'(x) + \dots + f_n'(x)$, $(\forall) x \in J$, $(\forall) n \geq 0$

S_n converge simplu pe J.

$S_n \xrightarrow[J]{c.s.} \sum_{n=0}^{\infty} f_n \overset{not}{=} S$

$S_n' \xrightarrow[J]{c.u.} \sum_{n=0}^{\infty} f_n' \overset{not}{=} T$

At. S e derivabilă și $S'(x) = T(x)$, $(\forall) x \in J$.

$\left(\sum_{n=0}^{\infty} f_n\right)'(x) = \sum_{n=0}^{\infty} f_n'(x)$

Def: Fie $f_n : A \to \mathbb{C}$, $n \geq q$, şir de funcţ.

$(f_n)_{n \geq q}$ s.n. uniform CAUCHY pe A dacă $(\forall)\ \varepsilon > 0$, $(\exists)\ n_\varepsilon \geq q$,

a.î. $(\forall)\ n \geq n_\varepsilon$, $(\forall)\ p \in \mathbb{N}^*$, $(\forall)\ x \in A$, avem:

$$|f_{n+p}(x) - f_n(x)| < \varepsilon$$

Teorema: $(f_n)_{n \geq q}$ este uniform CAUCHY pe $A \iff$

$\iff (f_n)_n$ este uniform convergent pe A.

Fie $\varepsilon > 0$, $(\exists)\ n_\varepsilon \geq q$ a.î. $(\forall)\ n \geq n_\varepsilon$, $(\forall) p \in \mathbb{N}^*$, $(\forall) x \in A$

avem: $|f_{n+p}(x) - f_n(x)| < \frac{\varepsilon}{2} < \varepsilon \Rightarrow$

$\Rightarrow (f_n(x))_n$ este şir (numeric) CAUCHY $((\forall) x \in A)$.

Deci $(\forall) x \in A \Rightarrow f_n(x)$ este convergent.

Fie $f(x) \overset{not}{=} \lim\limits_{n \to \infty} f_n(x) \Rightarrow \lim\limits_{p \to \infty} f_{n+p}(x) = f(x)$

$f_n \overset{S.}{\underset{A}{\to}} f$.

$$\lim_{p \to \infty} |f_{n+p}(x) - f_n(x)| \leq \frac{\varepsilon}{2}$$

$$\| \quad |f(x) - f_n(x)| \leq \frac{\varepsilon}{2} < \varepsilon \qquad (\forall)\ n \geq n_\varepsilon,\ (\forall)\ x \in A$$

$$\Rightarrow f_n \overset{U.}{\underset{A}{\to}} f.$$

(T) Crit. CAUCHY de uniform convergență pt. serii de funcții.

Fie $f_n : A \longrightarrow \mathbb{C}$ funcțiile $(\forall) n \geq p \in \mathbb{N}$

Atunci:

$\sum_{n \geq p} f_n$ este uniform conv. pe $A \iff (\forall) \varepsilon > 0$,

$(\exists) n \geq p$ a.î. $(\forall) n \geq n_\varepsilon$, $(\forall) p \in \mathbb{N}^*$ și $(\forall) x \in A$ avem

$$\left| f_{n+1}(x) + f_{n+2}(x) + \dots + f_{n+p}(x) \right| < \varepsilon$$

$$\left| \sum_{k=n+1}^{n+p} f_k(x) \right|$$

Dem: $S_n = f_p(x) + \dots + f_n(x)$

$\sum_{n \geq p} f_n$ u.c. $\iff (S_n)_n$ e Uniform. Conv. $\iff S_n$ uniform CAUCHY

$\iff (\forall) \varepsilon > 0$, $(\exists) n_\varepsilon \in \mathbb{N}$, $n_\varepsilon \geq p$ a.î. $(\forall) n \geq n_\varepsilon$, $(\forall) p \in \mathbb{N}^*$

$\forall / x \in A$ avem $|S_{n+p}(x) - S_n(x)| < \varepsilon$

$$\left| \sum_{k=n+1}^{n+p} f_k(x) \right|$$

Prop: $f_n : A \to \mathbb{C}$, $n \geq q$.

Dacă $\sum_{n \geq q} |f_n x|$ este U.C. pe $A \Rightarrow \sum_{n \geq q} f_n(x)$ e U.C. pe A.

Dem: $(\forall)\ \varepsilon > 0$, $(\exists)\ n_\varepsilon \geq p$ a.î. $(\forall) n \geq n_\varepsilon$, $(\forall) p \in \mathbb{N}^*$, $(\forall) x \in A$ avem:

$$|f_{n+1}(x)| + \ldots + |f_{n+p}(x)| < \varepsilon$$

$$\Vert$$

$$|f_{n+1}(x) + \ldots + f_{n+p} x| \quad \text{crit. Cauchy}$$

Teoremă: $(f_n)_{n \geq q}$, $(g_n)_{n \geq q}$, ~~$f_n, g_n : A \Rightarrow$~~

$f_n : A \to \mathbb{C}$; $g_n : A \to \mathbb{R}$

Pp că $(\exists)\ N \geq q$ a.î. $(\forall) n \geq N$, $(\forall) x \in A$, avem:

$|f_n(x)| \leq g_n(x)$

Dacă $\sum_n g_n$ este U.C. pe $A \Rightarrow \sum_n f_n$ este U.C. pe A.

Dem: $\varepsilon > 0$, $(\exists)\ r_\varepsilon \geq N$ a.î. $(\forall) n \geq r_\varepsilon$, $(\forall) p \in \mathbb{N}^*$, $(\forall) x \in A$ avem $g_{n+1}(x) + \ldots + g_{n+p}(x) < \varepsilon$

$$\Vert$$

$$|f_{n+1}(x)| + \ldots + |f_{n+p}(x)|$$

$$\Vert$$

$$|f_{n+1}(x) + \ldots + f_{n+p}(x)|$$

Obs: Fie $(z_n)_{n\geq 0}$ șir de nr. complexe

Dacă $\sum_n z_n$ este convergentă $\Rightarrow \sum_n z_n$ este U.C. pe A.

$S_n = z_0 + \dots + z_n$

(T) Criteriul WEIERSTRASS de uniform convergență a șirilor de funcții

Fie $f_n : A \to \mathbb{C}$, $n \geq 0$.
Fie $(a_n)_n$ șir de nr. reale
Pp că $(\exists) N \geq 0$ a.î. $(\forall) n \geq N$, $(\forall) x \in A$ avem:
$|f_n(x)| \leq a_n$

Dacă $\sum_{n\geq 0} a_n$ este conv $\Rightarrow \sum_{n\geq 0} f_n$ este U.C. pe A.

(T) Criteriul DIRICHLET de uniform conv. pt. serii de funcț.

$f_n : A \to \mathbb{R}$, $g_n : A \to \mathbb{C}$, $n \geq 0$

- Pp. că $(\exists) N \geq 0 \in \mathbb{N}$, a.î. $(\forall) n \geq N$, $(\forall) x \in A$ avem:
$f_{n+1}(x) \leq f_n(x)$
- Pp că $f_n \xrightarrow[A]{u.c.} 0$
- Fie $S_n(x) = g_N(x) + \dots + g_n(x)$, $(\forall) x \in A$, $(\forall) N \leq n$
Pp că $(\exists) M > 0$, a.î. $(\forall) n \geq N$, $(\forall) x \in A \Rightarrow |S_n(x)| \leq M$.

Atunci: $\sum_{n\geq 0} f_n \cdot g_n$ este U.C. pe A.

Șiruri

(T) Criteriul ~~Cri~~ LEIBNIZ de uniform conv. a șirilor de funcț.

Fie $f: A \to \mathbb{C}$, $n \geq 0$

- Pp. că: $(\exists)\ N \geq 0$, a.î. $(\forall)\ n \geq N$, $(\forall)\ x \in A \Rightarrow f_{n+1}(x) \leq f_n(x)$
- Pp. că: $f_n \xrightarrow[A]{u.c.} 0$

Atunci $\sum_n (-1)^n f_n$ și $\sum_n (-1)^{n-1} f_n$ sunt uniform conv. pe A.

(T) Crit. ABEL de uniform conv. a șirilor de funcț.

$f_n: A \to \mathbb{R}$, $g_n: A \to \mathbb{C}$, $n \geq 0$

- Pp. că $(\exists)\ N \in \mathbb{N}$, $N \geq 0$. a.î. $(\forall)\ n \geq N$, $(\forall) x \in A \Rightarrow$
$\Rightarrow f_{n+1}(x) \leq f_n(x)$
- Pp. că $(\exists)\ M > 0$ a.î. $(\forall)\ n \geq N$, $(\forall)\ x \in A$ avem:
$|f_n(x)| \leq M$.
- Pp. că seria de funcții $\sum_{n \geq 0} g_n$ este U.C. pe A

Atunci $\sum_{n \geq 0} f_n \cdot g_n$ este U.C. pe A

$$\sum_{n\geq 1} (-1)^n \frac{(n^2+1)x}{x^2+n^4}$$

Stab. C.U. și C.S.

a) $x \in [0,1]$

b) $x \in [1,\infty)$

$$f_n(x) = (-1)^n \frac{(n^2+1)x}{x^2+n^4}$$

$$|f_n(x)| = |(-1)^n| \cdot \frac{(n^2+1)x}{n^2\left(n^2+\frac{x^2}{n^2}\right)}$$

$$\frac{|f_n(x)|}{\frac{1}{n^2}} = \frac{(n^4+n^2)x}{n^4+x^2} \to 1 \cdot x \neq 0 \Rightarrow \sum \frac{1}{n^2} \text{ conv} \Rightarrow$$

$$\Rightarrow \sum_n |f_n(x)| \text{ conv} \Rightarrow \sum_n f_n(x) \text{ este abs. conv} \Rightarrow \text{și conv.} \quad (\forall) x \geq 0$$

$\sum_n f_n$ converge simplu pe $(\forall) x \in [0,1]$ și $x \in [1,+\infty)$ deci pe $[0,\infty)$ converge simplu pe $(\forall) x \geq 0$

$$\Rightarrow f_n \xrightarrow[x\in[0,\infty)]{S.C.} 0$$

$$|f_n(x)| = \frac{(n^2+1)\cdot x}{x^2+n^4} = h_n(x)$$

$$h_n'(x) = \frac{(n^2+1)x^2+n^4 - x\cdot 2x}{(x^2+n^4)^2} = (n^2+1)\cdot\frac{n^4-x^2}{(x^2+n^4)^2}, \quad x = n^2$$

a) $0 \leq x \leq 1$

$n \geq 2$

x	0		1		n^2
$h_n'(x)$	$+$	$+$	$+$	$+$	0
$h_n(x)$		↗			$h_n(1)$

$\sup_{x\in[0,1]} h_n(x) = h_n(1) = \frac{n^2+1}{n^4+1}$

$\Leftarrow$ "$|f_n(x)|$"

$|f_n(x)|$ $(\forall) n\geq 2$, $(\forall) x\in[0,1]$

$$|f_n(x)| \leq \frac{n^2+1}{n^4+1} = a_n$$

$$\frac{a_n}{\frac{1}{n^2}} = \frac{n^2+n^4}{n^2+1} \to 1 \neq \infty$$

$\sum_n \frac{1}{n^2}$ conv $\Rightarrow$

$\Rightarrow \sum_n a_n$ conv

Weierstrass $\Longrightarrow \sum_n f_n$ e. U.C. pe $[0,1]$

(2) $n\geq 2$

x	1		n^2		∞
$h_n'(x)$	$+$	$++$	0	$---$	
$h_n(x)$		$\nearrow$	$h_n(n^2)$	$\searrow$	

$\sup_{x\in[1,\infty)} h_n(x) = h_n(n^2) =$

$= \frac{n^4+n^2}{n^4+n^4} = \frac{n^2+1}{2n^2}$

$\sup_{x\in[1,\infty)} |f_n(x)-0| = \frac{n^4+n^2}{n^4+n^4} = \frac{n^2+1}{2n^2} \to \frac{1}{2}$ (1)

Pp că $\sum_n f_n$ e uniform conv pe $[1,\infty) \Rightarrow f_n \xrightarrow[[1,\infty)]{U.C.} 0 \Leftrightarrow$

$\Leftrightarrow \sup_{x\in[1,\infty)} |f_n(x)-0| \to 0$ (2)

Din (1) și (2) $\Rightarrow \sum_n f_n$ nu e U.C. pe $[1,\infty)$

Aplicații (pct a)

$$|f_n(x)| = \frac{(n^2+1)x}{x^2+n^4} \leq \frac{(n^2+1)\cdot 1}{x^2+n^4} \leq \frac{n^2+1}{n^4}$$

$$|f_n(x)| \leq \underbrace{\frac{n^2+1}{n^4}}_{=\alpha_n>0}, \ (\forall) n \geq 1, (\forall) x \in [0,1]$$

$$\frac{\alpha_n}{\frac{1}{n^2}} = \frac{n^2+1}{n^2} \longrightarrow 1 \neq \infty$$

$\sum_n \frac{1}{n^2}$ conv $\Rightarrow \sum_n \alpha_n$ conv

$|f_n(x)| \leq \alpha_n$, $(\forall) x \in [0,1]$, $(\forall) n \geq 1$

$\sum_n \alpha_n$ conv

Weier. $\Rightarrow \sum_n f_n$ e U.C. pe $[0,1] \Rightarrow$

$\Rightarrow \sum_n f_n$ S.C. pe $[0,1]$

Ex: $\sum_{n\geq 1} (-1)^n \cdot \frac{(n^2+1)x}{n^3+x^2}$, $x \in [0,\infty)$

a) Arătați că seria converge simplu pe $[0,\infty)$

b) Fie $S(x) = \sum_{n=1}^{\infty} (-1)^n \cdot \frac{(n^2+1)x}{n^3+x^2}$, $(\forall) x \in [0,+\infty)$

Este S continuă? Dar derivabilă?

Se poate deriva termen cu termen?

$$f_n(x) = (-1)^n \underbrace{\frac{(n^2+1)x}{n^3+x^2}}_{=h_n(x)}, \ x \geq 0$$

(a) $f_n(0) = 0 \Rightarrow \sum_{n\geq 1} f_n(0)$ conv.

$x > 0$ $x \downarrow x_0 +$

$f_n(x) = (-1)^n h_n(x)$, $h_n(x) \xrightarrow{n\to\infty} 0$

$\varphi_x(y) = \dfrac{(y^2+1)x}{y^3+x^2}$, $y \geq 1$

$\varphi'_x(y) = x \cdot \dfrac{2y(y^3+x^2) - (y^2+1)\cdot 3y^2}{(y^3+x^2)^2} =$

$= \dfrac{xy(2x^2 - 3y - y^3}{(y^3+x^2)^2}$ $x < y$ si $y > 1$

Daca $y > x^2+1 \Rightarrow \varphi'_x(y) < 0$

$\varphi_x(y)$ e desc. pe (x^2+1, ∞)

$(\forall)\ n > x^2+1$, $n = [x^2] + 2 > x^2$

$n+1 > n > x^2+1$

$\Downarrow$

$\varphi_x(n+1) < \varphi_x(n)$, $\varphi_x(n) = h_n(x)$

$h_{n+1}(x) < h_n(x)$, $(\forall) n > x^2+1$

$h_n(x) \xrightarrow[n]{} 0$ $\Rightarrow \sum_n (-1)^n h_n(x)$ convergentă $(\forall)\ x > 0$

6) $f_n(x) = (-1)^n \cdot h_n(x)$

$f_n'(x) = (-1)^n \cdot h_n'(x) = (-1)^n \cdot (n^2+1) \cdot \dfrac{x^2 + n^3 - x \cdot 2x}{(x^2+n^3)^2}$

$f_n'(x) = (-1)^n \dfrac{(n^2+1)(n^3-x^2)}{(n^3+x^2)^2} \cdot (-1)^n \cdot \dfrac{n^5}{(n^3+x^2)^2} + (-1)^n \dfrac{(n^3 - x^2 - n^2x^2)}{(n^3+x^2)^2}$

$f_n'(x) = v_n(x) + \psi_n(x)$

$|\psi_n(x)| = \dfrac{|n^3 - x^2 - n^2x^2|}{(n^3+x^2)^2}$

Fie $a > 0$. Arătăm că $\sum_n \psi_n$ e uniform conv. pe $[0, a)$

$0 \le x < a$

$|\psi_n(x)| \le \dfrac{n^3 + |-x^2| + |-n^2x^2|}{(n^3+x^2)^2} = \dfrac{n^3 + x^2 + n^2 \cdot x^2}{(n^3+x^2)^2} \le$

$\le \dfrac{n^3 + (1+n^2)x^2}{n^6}$ $\quad x^2 < a^2$ $\quad \le \dfrac{n^3 + (1+n^2)a^2}{n^6}$ $\quad n^3 + x^2 \ge n^3$

$(\forall)\ x \in [0, a)$, $(\forall)\ n \ge 1$

$|\psi_n(x)| \le \dfrac{n^3 + (1+n^2)a^2}{n^6} = \alpha_n$

$$\frac{\alpha_n}{\frac{1}{n^3}} = \frac{n^6 + a^2 n^5 + a^2 n^3}{n^6} \to 1 \neq \infty$$

$\sum_n \frac{1}{n^3}$ conv $\Rightarrow \sum_n \alpha_n$ conv.

$|\psi_n(x)| \leq \alpha_n$, $(\forall)\, n \geq 1$, $(\forall)\, x \in [0, a)$

$\Rightarrow \sum_n \psi_n$ e. U. C. pe $[0, a)$

$f_n'(x) = v_n(x) + \psi_n(x)$, $(\forall)\, x \geq 0$, $(\forall)\, n \geq 1$

$v_n(x) = (-1)^4 \cdot \frac{n^5}{(n^3 + x^2)^2} = \beta_n(x)$

$\beta = \frac{n^5}{(n^3+x^2)^2} \leq \frac{n^5}{n^6} \leq \frac{1}{n} \to 0 \Rightarrow \beta \xrightarrow{u.c.} 0$

$w(y) = \frac{y^5}{(y^2+x^2)^2} \Rightarrow w'(y) = \frac{5y^4}{(\dots)^2} + \frac{y^5(2)}{(\dots)^2} \cdot 3y^2 =$

$= \frac{y^4(5y^3 + 5x^2 - 6y^3)}{(\dots)^3} = \frac{y^4(5x^2 - y)}{(\dots)^3} \Rightarrow$

$\Rightarrow y \geq \sqrt[3]{5x^2}$

$0 \le x \le a$, $y \ge \sqrt[3]{5a^2}$ $\quad W'(y) \le 0$

$\Updownarrow$

W descr pe $[\sqrt[3]{5a^2}, \infty)$

$(\forall)\ n \ge \sqrt[3]{5a^2}$

$n+1 > n \ge \sqrt[3]{5a^2}$

$\Updownarrow$

$W_{(n+1)} \le W_n$

$W_{(n+1)} \le W_{(n)}$

$\Updownarrow$

$\beta_{(n+1)}(x) \le \beta_n(x)$

$$|\beta_n(x)| = \frac{n^5}{(n^3+x^2)^2} \le \frac{n^5}{(n^3)^2} = \frac{1}{n} \to 0 \quad (\forall) n \ge 1, (\forall) x \ge 0$$

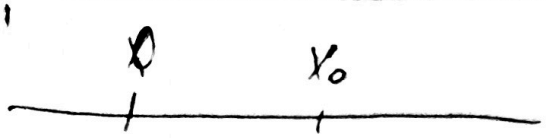
$\Rightarrow \beta_n \xrightarrow[[0,a)]{u.c.} 0$

$\underbrace{\quad\quad}_{=\alpha_n(x)}$

Leib $\Rightarrow \sum_n (-1)^n \beta_n(x)$ u. c. pe $[0,a)$

$f'_n(x) = \underset{u.c.}{\alpha_n(x)} + \underset{u.c.}{\psi_n(x)} \Rightarrow \sum_n f'_n$ u. c. pe $[0,a)$

$\sum_n f_n$ s. c. pe $[0,\infty)$



Fie $x_0 \geq 0$. Fie $a = x_0 + 1$, $x_0 \in [0, a)$

T. de derivare termen cu termen pe $[0, a)$ $\Rightarrow$

S e derivabilă pe $[0, a)$ și $S'(x) = \sum_{n=1}^{\infty} f_n'(x)$ $(\forall) x \in [0, a)$

S e derivabilă în x_0 și $S'(x_0) = \sum_{n=1}^{\infty} f_n'(x_0)$ $(\forall) x_0 \geq 0$

$(\forall)\, x \geq 0$, S e derivabilă în x și $S'(x) = \sum_{n=1}^{\infty} f_n'(x)$, $(\forall) x \geq 0$

S e derivabilă pe $[0, \infty) \Rightarrow S$ e continuă $\Rightarrow$

$\Rightarrow$ Se poate deriva termen cu termen.

Ex: $\sum_{n\geq 1} \underbrace{(-1)^n \frac{n^5}{(n^3+x^2)^2}}_{=u_n(x)}$, $x \geq 0$

! Seria este U.C. pe $[0,\infty)$

$u_n(x) = \frac{(-1)^n}{n} \cdot \left(\frac{n^3}{n^3+x^2}\right)^2 = \underbrace{\frac{(-1)^{n\cdot 1}}{n}}_{f_n(x)} \cdot \underbrace{\left(-\left(\frac{n^3}{n^3+x^2}\right)^2\right)}_{g_n(x)}$

$x \geq 0$
$n \geq 1$

ABEL

$\sum_n \frac{(-1)^{n\cdot 1}}{n}$ este U.C. pe $[0,\infty)$

$|f_n(x)| = \left(\frac{n^3}{n^3+x^2}\right)^2 \leq 1$, $(\forall)\, x \in [0,\infty)$

$n^3 \leq x^3 + x^2$

$f_{n+1}(x) \leq f_n(x) \iff -f_{n+1}(x) \geq -f_n(x) \iff$

$\left(\frac{(n+1)^3}{(n+1)^3+x^2}\right)^2 \geq \left(\frac{n^3}{n^3+x^2}\right)^2 \iff \frac{(n+1)^3}{(n+1)^3+x^2} \geq \frac{n^3}{n^3+x^2} \iff$

$\iff n^3(n+1)^3 + (n+1)^3 x^2 \geq n^3(n+1)^3 + n^3 x^2 \iff$

$\iff (n+1)^3 \cdot x^2 \geq n^3 x^2$

$(\forall)\, x \in [0,\infty)$, $n \geq 1$

ABEL
$\Rightarrow$ $\sum_{n\geq 1} f_n(x) \cdot g_n(x)$ Uniform Conv pe $[0,\infty) \iff$

$\iff \sum_n \frac{(-1)^{n\cdot 1}}{n} \cdot \frac{-n^6}{(n^3+x^2)^2}$ U.C. pe $[0,\infty)$

(1) $f_n : J \to \mathbb{C}$, f_n continue

$f_n \xrightarrow[J]{u.c.} f : J \to \mathbb{C}$

At: f e continue pe J

Dem: Fie $\varepsilon > 0$; $(\exists)\, n_\varepsilon \geq 1$ a.i. $(\forall) n \geq n_\varepsilon$, $(\forall) x \in J$

$\Rightarrow |f_n(x) - f(x)| \leq \frac{\varepsilon}{3}$ (1)

Fie $a \in J$. f_{n_ε} e cont in a:

$(\exists)\, \delta_\varepsilon > 0$ a.i. $(\forall) x \in J$ cu $|x - a| \leq \delta_\varepsilon \Rightarrow$

$\Rightarrow |f_{n_\varepsilon}(x) - f_{n_\varepsilon}(a)| < \frac{\varepsilon}{3}$ (2)

$$|f(x) - f(a)| = |f(x) - f_{n_\varepsilon}(x) + f_{n_\varepsilon}(x) - f_{n_\varepsilon}(a) + f_{n_\varepsilon}(a) - f(a)| \leq$$

$$\leq \underbrace{|f(x) - f_{n_\varepsilon}(x)|}_{\text{Conf. (1)} < \varepsilon/3} + \underbrace{|f_{n_\varepsilon}(x) - f_{n_\varepsilon}(a)|}_{\text{Conf. (2)} < \varepsilon/3} + \underbrace{|f_{n_\varepsilon}(a) - f(a)|}_{\text{Conf. (1)} < \varepsilon/3} < \varepsilon$$

$(\forall) x \in J$ cu $|x - a| < \delta_\varepsilon \Rightarrow f$ continuă în a.

(T) $f_n : [a,b] \to \mathbb{C}$ continuă

$f_n \xrightarrow[[a,b]]{u.c.} f : [a,b] \to \mathbb{C}$

Af. $\int_a^b f_n(x)\,dx$ e conv $\Rightarrow \int_a^b f_n(x)\,dx \to \int_a^b f(x)\,dx$

Fie $\varepsilon > 0$. $(\exists)\ n_\varepsilon$ a.î. $(\forall)\ n \geq n_\varepsilon$, $(\forall)\ x \in [a,b]$

$|f_n(x) - f(x)| < \dfrac{\varepsilon}{2(b-a)}$

$\left| \int_a^b f_n(x)\,dx - \int_a^b f(x) \right| = \left| \int_a^b (f_n(x) - f(x))\,dx \right| \leq \int_a^b |f_n(x) - f(x)|\,dx \leq$

$\leq \int_a^b \dfrac{\varepsilon}{2(b-a)}\,dx = \dfrac{\varepsilon}{2(b-a)} \cdot (b-a) = \dfrac{\varepsilon}{2} < \varepsilon$

(T) $f_n : J \to \mathbb{C}$ deriv.

$f_n \xrightarrow[J]{S} f : J \to \mathbb{C}$

$f_n' \xrightarrow[J]{u.} g : J \to \mathbb{C}$

Af: f e derivabilă și $f'(x) = g(x)$

Dem: în ipoteza suplimentară că f_n sunt de cl. C^1, adică f_n derivabilă și f_n' cont. pe J.

Fie $a \in J$, $x \in J$, $x \neq a$

I precedentă pt. $\varphi_n(x) = f_n'(x)$

pe $[a,x]$ sau $[x,a]$

$\int_a^x \varphi_n(t)\,dt \to \int_a^x g(t)\,dt$

$\int_a^x f_n'(t)\,dt = f_n(x) - f_n(a) \to f(x) - f(a)$

$f_n(x) - f_a^x$ $f_n(x) - f(a) = \int_a^x g(t)\,dt$

$\varphi_n = \underbrace{f_n'}_{\text{cont.}} \xrightarrow{u} g \Rightarrow g$ cont. pe J.

$(J)\ G: J \to \mathbb{C}$, $G'(x) = g(x)$, $\forall x$

$f(x) = G(x) - G(a) + f(a)$

G deriv.

f deriv. $\Rightarrow f$ deriv.

$f'(x) = G'(x) = g(x)$

(T) Cut ABEL de uniform convergență.

$f_n: A \to \mathbb{R}$, $g_n: A \to \mathbb{C}$, $n \geq p$

- $\exists\, N \in \mathbb{N}$, $N \geq p$ a.î. $(\forall)\, n \geq N$, $\forall x \in A \Rightarrow$

$\Rightarrow f_{n+1}(x) \leq f_n(x)$

$(\exists)\, M > 0$ a.î. $|f_n(x)| \leq M$, $(\forall)\, n \geq N$, $\forall x \in A$

- $\sum_{n \geq p} g_n$ e unif conv. $\Rightarrow \sum_{n \geq p} f_n g_n$ e uniform conv. pe A

Dem $\sum_{n \geq N} g_n$ e uniform conv.

$S_n(x) = g_N(x) + g_{N+1}(x) + \dots + g_n(x)$, $n \geq N$

$S(x) = \sum_{n=N}^{\infty} g_n(x)$, $S_n \xrightarrow[A]{u} S \iff \underbrace{(S_n - S)}_{\tilde{S}} \xrightarrow[A]{u} 0$

$\tilde{S}(x) = S_n(x) - S(x)$, $\tilde{S}_n(x) - \tilde{S}_{n-1}(x) = g_n(x)$, $\forall n \geq N+1$

$(\forall)\ n \geq N,\ (\forall)\ x \in A,\ (\forall)\ p \in \mathbb{N}^*$

$$\sum_{k=n+1}^{n+p} f_k(x) \cdot g_k(x) = \sum_{k=n+1}^{n+p} \left(\tilde{S}_k(x) - \tilde{S}_{k-1}(x) \right) \cdot f_k(x)$$

$$= \sum_{k=n+1}^{n+p} f_k(x) \cdot \tilde{S}_k(x) - \sum_{k=n+1}^{n+p} f_k(x) \cdot \tilde{S}_{k-1}(x)$$

$$= \sum_{k=n+1}^{n+p} f_k(x)\, \tilde{S}_k x - \sum_{k=n}^{n+p-1} f_{k+1}(x) \cdot \tilde{S}_k(x)$$

$n \leq k-1 = j \leq n+p-1$

$$= \sum_{k=n+1}^{n+p-1} \left[f_k(x) - f_{k+1}(x) \right] \tilde{S}_k(x) - f_{n+1}(x) \cdot \tilde{S}_n(x) + f_{n+p}(x) \cdot \tilde{S}_{n+p}(x)$$

$$\left| \sum_{k=n+1}^{n+p} f_k(x)\, g_k(x) \right| \leq \sum_{k=n}^{n+p-1} \left| f_k(x) - f_{k+1}(x) \right| \cdot \left| \tilde{S}_k(x) \right| + \left| f_{n+1}(x) \right| \cdot \left| \tilde{S}_n(x) \right| +$$

$$+ \left| f_{n+p}(x) \right| \cdot \left| \tilde{S}_{n+p}(x) \right| \leq \sum_{k=p}^{n+p-1} \underbrace{\left| f_k(x) - f_{k+1}(x) \right|}_{\geq 0} \cdot \left| \tilde{S}_k(x) \right| +$$

$k \geq n \geq N$

$$+ M \left| \tilde{S}_n(x) \right| + M \left| \tilde{S}_{n+p}(x) \right|$$

$\tilde{S}_n \xrightarrow[A]{u} 0$

$(\exists)\ n_\varepsilon \geq N$ a.î. $(\forall)\ n \geq n_\varepsilon,\ (\forall)\ x \in A$

$$\left| \tilde{S}_n(x) \right| < \frac{\varepsilon}{5M} \qquad k \geq n \geq n_\varepsilon$$

$$\leq \sum_{k=n}^{n+p} \left(f_k(x) - f_{k+1}(x) \right) \cdot \frac{\varepsilon}{5M} + M \cdot \frac{\varepsilon}{5M} + M \cdot \frac{\varepsilon}{5M} =$$

$f_n + f_{n+1} + \dots + f_{n+p}$

$- f_{n+1} - f_{n+2} - \dots - f_{n+p} - f_{n+p+1}$

$$= \frac{\varepsilon}{5M} \cdot f_n(x) - \frac{\varepsilon}{5M} \cdot f_{n+p+1}(x) + \frac{2}{5}\varepsilon$$

$\leq |f_n(x)|$

$\leq \frac{\varepsilon}{5M} \cdot M + \frac{\varepsilon}{5M} \cdot M + \frac{2}{5}\varepsilon = \frac{4}{5}\varepsilon < \varepsilon$

Serii de puteri

$\sum_{n\geq 0} a_n z^n$, a_n şir de nr complexe

$z \in \mathbb{C}$

$S_n(z) = a_0 + a_1 z + \dots + a_n z^n$

$S_n(0) = a_0 \longrightarrow a_0$

$|a_0| + |a_1 z| + \dots + |a_n z^n| = \sum_{n\geq 0} |a_n z^n|$

pt $z = 0$: $|a_0| \to |a_0|$

pt $z = 0$ seria $\sum_{n\geq 0} a_n z^n$ este absolut convergentă

$W \overset{not}{=} \{ r \geq 0 \mid \sum_{n\geq 0} |a_n| r^n$ este conv. $\}$

$W \neq \emptyset$

W

Def: $R = \sup\{ r \geq 0 \mid \sum_{n\geq 0} |a_n| r^n$ este conv. $\}$

$R = [0, \infty]$

$\Rightarrow$ raza de convergenţă a seriei de puteri $\sum_{n\geq 0} a_n z^n$

Lemă:

Fie $z_0 \in \mathbb{C}$, $z_0 \neq 0$

Pp că $\sum_{n\geq 0} a_n z_0^n$ este conv.

Atunci $\forall z \in \mathbb{C}$, $|z| \leq |z_0|$, avem $\sum_{n\geq 0} a_n z^n$ este absolut conv.

Dem: $\Rightarrow a_n z_0^n \to 0$ (conv. şi mărginit)

$(\exists) M > 0$ a.î. $|a_n \cdot z_0^n| \leq M$, $(\forall) n \geq 0$

$|a_n z^n| = |a_n \cdot z_0^n| \cdot \left|\left(\frac{z}{z_0}\right)^n\right| \leq M \left|\frac{z}{z_0}\right|^n$

$0 \leq |a_n z^n| \leq M \left|\frac{z}{z_0}\right|^n, (\forall) n \geq 0$ (1)

$|z| < |z_0| \Rightarrow \left|\frac{z}{z_0}\right| = \frac{|z|}{|z_0|} \leq 1 \Rightarrow \sum_n \left|\frac{z}{z_0}\right|^n$ conv $\Rightarrow$

$\Rightarrow \sum_n M \cdot \left|\frac{z}{z_0}\right|^n$ conv. (2)

Din (1) + (2) $\Rightarrow \sum_n a_n z_0^n$ abs. conv.

(T): Fie R - raza de convergență a seriei $\sum_{n=0} a_n z^n$

Atunci 1) Dacă $|z| < R \Rightarrow \sum_{n \geq 0} a_n z^n$ este absolut conv.
(dacă $R=0$ acest caz nu apare)

2) Dacă $|z| > R \Rightarrow \sum_{n \geq 0} a_n z^n$ este divergentă
(Dacă $R = \infty$ acest caz nu apare)

Dem: 1) Pp. că $(\exists)\ v \in \mathbb{C}$, $|v| < R$ a.î. $\sum_n |a_n v^n|$ div.

Arătăm că $(\forall)\ r \in W \Rightarrow r \leq |v|$

(Pp că $(\exists)\ r_0 \in W$ cu $r_0 \geq |v| \Rightarrow \sum_{n \geq 0} |a_n r_0^n|$ conv $\Rightarrow \sum_n a_n r_0^n$ conv

$\Rightarrow \sum_n a_n v^n$ abs conv $\Rightarrow$ FALS.)

$(\forall) r \in W \Rightarrow r \leq |v| \Rightarrow |v|$ e majorant pt. W $\Rightarrow R \leq |v|$
$R = \sup W$ contradicție

$\Rightarrow \sum_n |a_n v^n|$ conv

a) $|z| > R \Rightarrow \sum_n a_n z^n$ div

Pp. că $\sum_n a_n z^n$ conv.

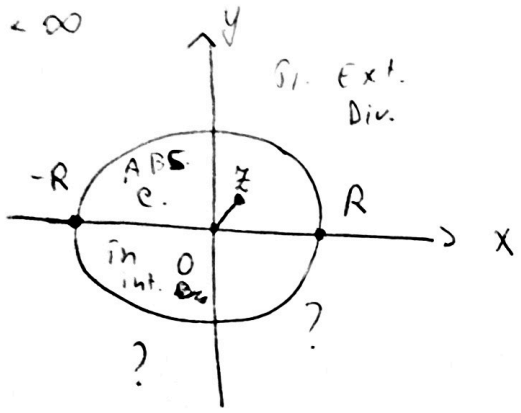
Fie $r \geq 0$, a.î. $R < r < |z|$, $r = \frac{R+|z|}{2}$

LEMA $\Rightarrow \sum_n a_n r^n$ abs. conv.

$\sum_n |a_n| r^n$ conv $\Rightarrow r \in W$

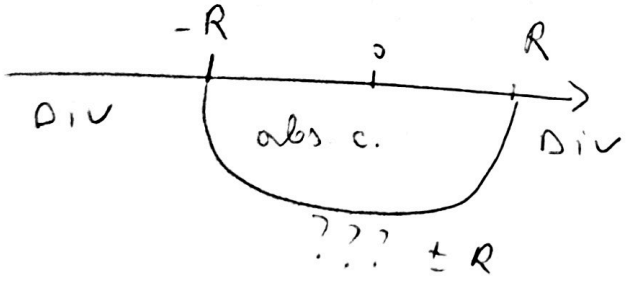
R sup W $\Rightarrow R \geq r$ FALS.

$0 < R < \infty$



$x \in \mathbb{R}$, $\sum_{n \geq 0} a_n x^n$

$0 < R < \infty$



Dacă $R = \infty \Rightarrow \sum_n a_n z^n$ abs c.

Dacă $R = 0 \Rightarrow$ $z \neq 0$ $\sum_n a_n z^n$ div.

Obs: Fie $\alpha \in \mathbb{R}$, $\alpha > 0$. Pp că

1) $(\forall)\ x \in (0, \alpha) \Rightarrow \sum_{n \geq 0} a_n x^n$ este conv.

2) $(\forall)\ x > \alpha \Rightarrow \sum_n a_n x^n$ este div

$\Rightarrow R = \alpha$ este unica valoare cu proprietatea că seria $\sum_{n \geq 0} a_n z^n$

Teorema

$\sum_{n\geq 0} a_n z^n$, $a_n \in \mathbb{C}$

$z \in \mathbb{C}$

Ap. că $(\exists) \lim_{n\to\infty} \sqrt[n]{|a_n|} \overset{not.}{=} L \in [0, \infty]$

Fie R - raza de conv.

At: $R = \frac{1}{L}$ $\left(\frac{1}{0} = \infty, \frac{1}{\infty} = 0\right)$

Dem: pt. $z = 0 \Rightarrow \sum_n a_n z^n$ abs. c.

Fie $z \in \mathbb{C}$, $(z \neq 0)$

$\sum_n |a_n z^n|$, $|a_n| \cdot |z|^n$

$\sqrt[n]{|a_n z^n|} = \sqrt[n]{|a_n|} \cdot |z| \to L \cdot |z|$

$\star$ $(\frac{1}{1})$: $0 < L < \infty$

Dem: $|z| \cdot L < 1$, $|z| < \frac{1}{L}$

- pt. $|z| < \frac{1}{L} \Rightarrow$ seria abs. conv.
  (și pt. $z = 0$)

- pt. $|z| > \frac{1}{L}$, $L \cdot |z| > 1 \Rightarrow |a_n z^n| \to \infty \neq 0$ $\star\star$

$\star\star$ $a_n z^n \to 0 \iff |a_n z^n| \to 0$

$|a_n z^n| \not\to 0 \Rightarrow a_n z^n \not\to 0$

$\Rightarrow \sum_n a_n z^n$ Div.

- pt $|z| < \frac{1}{L} \Rightarrow \sum_n a_n z^n$ abs. conv.
- pt $|z| > \frac{1}{L} \Rightarrow \sum_n a_n z^n$ Div.

Ⅱ $L = 0$

$\sqrt[n]{|a_n z^n|} \to 0 < 1 \Rightarrow \sum_n |a_n z^n|$ abs. conv. $\Rightarrow R = \infty$

(P) Dacă $a_n > 0$ și $\frac{a_{n+1}}{a_n} \to L \Rightarrow \sqrt[n]{a_n} \to L$

Teor $a_n = 0$, $(\forall)$ $n \geq p \geq 0$, $p \in \mathbb{N}$

Pp că $(\exists) \lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} \overset{not}{=} L \in [0, \infty]$

$\Rightarrow$ raza de conv. a seriei este $R = \frac{1}{L}$

Ex: 1) $\sum_{n\geq 0} z^n$

pt. $|z| < 1 \Rightarrow$ s. c.
pt. $|z| \geq 1 \Rightarrow$ s. D. $\Big| \Rightarrow R = 1$ pt. $|z| = R = 1 \Rightarrow$ s. D.

$a_n = 1$

$\left(\sqrt[n]{|a_n|} = 1 \to \underset{L}{1} \Rightarrow R = \frac{1}{L} = 1\right)$

Ex 2 $\sum_{n\geq 1} \frac{1}{n^2} \cdot z^n$

$a_n = \frac{1}{n^2}$, $n \geq 1$

$\frac{|a_{n+1}|}{|a_n|} = \frac{n^2}{(n+1)^2} = \left(\frac{n}{n+1}\right)^2 \longrightarrow 1 \Rightarrow L$

$R = \frac{1}{L} = 1$ pt. $|z| < 1 = R$, seria e abs. c.

pt. $|z| > 1 \Rightarrow$ seria div

pt $|z| = 1 \Rightarrow R$

$\left|\frac{z^n}{n^2}\right| = \frac{|z|^n}{n^2} = \frac{1}{n^2}$, $\sum \frac{1}{n^2}$ conv.

Seria este abs. conv.

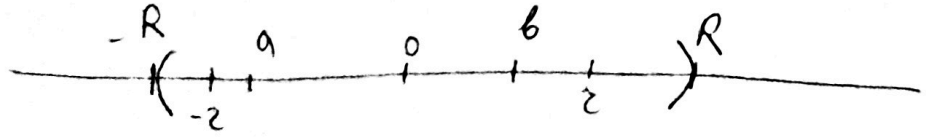
(3) Temă: $\sum_{n\geq 0} n! \cdot z^n$ are tatî cõ $R = 0$

Prop: $\sum_{n\geq 0} a_n x^n$ - are raza de conv R

Pp. $R \neq 0$. At. $(\forall)\ [a, b] \subseteq (-R, R)$

seria este un fsm conv pe $[a, b]$

$-\infty \leq -R < a < b < R \leq \infty$



$(\exists)\ z < R$ a.î. $[a,b] \subseteq [-z, z] \subseteq (-R, R)$

$z = \max\{|a|, |b|\}$

$\sum_n a_n z^n$ abs. c. $\Rightarrow \sum_n |a_n z^n|$ conv. (*)

$(\forall) x \in [a,b] \subseteq [-z, z]$, $|x| \le z$

$|a_n x^n| = |a_n| \cdot |x|^n \le |a_n| \cdot z^n = |a_n z^n|$

$(\forall) x \in [a,b]$, $(\forall) n \ge 0$

$|a_n x^n| \le |a_n z^n|$ (**)

(**), (*) $\Rightarrow$ $\sum_{n\ge 0} a_n x^n$ este uniform conv. pe $[a,b]$

Prop. a) Seriile $\sum_{n\ge 0} a_n x^n$ și $\sum_{n\ge 1} n a_n x^{n-1}$ au aceeași rază de conv.

b) Seriile $\sum_{n\ge 0} a_n x^n$ și $\sum_{n\ge 0} \frac{a_n}{a_{n+1}} x^{n+1}$ au aceeași rază de conv.

(b) rezultă din (a)

a) în ipoteza că ($\exists$) $\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} = L$

$\sum_n a_n x^n$ are raza $\frac{1}{L}$

$\sum_{n\ge 0} (n+1) a_{n+1} x^n$; $\frac{|n+2|\, a_{n+2}}{|(n+1)\, a_{n+1}|} = \frac{n+2}{n+1} \cdot \frac{|a_{n+2}|}{|a_{n+1}|} \to L$ ($\frac{n+2}{n+1} \to 1$, $\frac{|a_{n+2}|}{|a_{n+1}|} \to L$)

$\uparrow$ are raza $\frac{1}{L}$

Teorema: Fie $R \neq 0$
R raza de conv a seriei $\sum_{n\geq 0} a_n x^n$

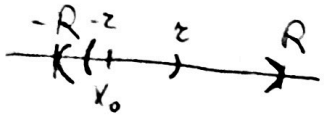
Fie $S: (-R, R) \to \mathbb{R}$, $S(x) = \sum_{n=0}^{\infty} a_n x^n$, $(\forall)\ x \in (-R, R)$

At: S este indefinit derivabilă pe $(-R, R)$

și $S'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$ $(\forall) x \in (-R, R)$

(se poate deriva termen cu termen pe $(-R, R)$)

Dem: $f_n(x) = a_n x^n$, $f_n'(x) = n a_n x^{n-1}$



Fie $x_0 \in (-R, R)$ $(\exists)\ r > 0$, $r < R$ a.î. $x_0 \in (-r, r)$, $|x_0| < r$

$(-r, r) \subseteq (-R, R) \Rightarrow \sum_n f_n$ conv. simplu pe $(-r, r)$

$\sum_n \underbrace{n a_n x^{n-1}}_{f_n'(x)}$ are raza de conv $R \Longrightarrow \sum_n f_n'$ uniform conv. pe $(-R, R)$

$\Rightarrow S$ este derivabilă pe $(-r, r)$ și $S'(x) = \sum_{n=1}^{\infty} f_n'(x)$, $(\forall) x \in (-r, r)$
sumă $\quad x_0 \in$

$\Rightarrow S$ e derivabilă în x_0 și $S'(x_0) = \sum_{n=1}^{\infty} n a_n x_0^{n-1}$ $(\forall) x_0 \in (-R, R)$

Teorema: $\sum_{n\geq 0} a_n x^n$, R - raza de conv.

Pp $R \neq 0$

Fie $[a, b] \subseteq (-R, R)$, $-\infty \leq -R < a < b < R \leq \infty$

At $\sum_{n\geq 0} a_n \frac{x^{n+1}}{n+1}\Big|_a^b$ este conv. și $\int_a^b \left(\sum_{n=0}^{\infty} a_n x^n\right) dx = \sum_{n=0}^{\infty} a_n \frac{x^{n+1}}{n+1}\Big|_a^b =$

$= \sum_{n=0}^{\infty} a_n \cdot \frac{b^{n+1} - a^{n+1}}{n+1}$

Teorema $\sum_{n\geq 0} a_n x^n$ are raza de conv R

Pp că $0 < R < \infty$. Pp. că $\sum_{n\geq 0} a_n R^n$ este conv $\Rightarrow$

$\Rightarrow$ i) $\sum_{n\geq 0} a_n x^n$ este uniform conv pe $[0, R]$

ii) $\sum_{n=0}^{\infty} a_n R^n = \lim_{\substack{x \to R \\ x < R}} \left(\sum_{n=0}^{\infty} a_n x^n \right)$

(suma seriei este continuă în R)

iii) seria $\sum_{n\geq 0} a_n \frac{R^{n+1}}{n+1}$ este conv și $\int_0^R \left(\sum_{n=0}^{\infty} a_n x^n \right) dx =$

$= \sum_{n=0}^{\infty} a_n \frac{x^{n+1}}{n+1} = \sum_{n=0}^{\infty} a_n \frac{R^{n+1}}{n+1}$

Dem:

i) $\sum_{n\geq 0} \underbrace{a_n R^n}_{g_n''(x)} \cdot \underbrace{\left(\frac{x}{R}\right)^n}_{f_n''(x)}$

$\sum_n \underbrace{a_n R^n}_{g_n''(x)}$ e uniform conv. pe $[0, R]$

$0 \leq x \leq R$

$|f_n(x)| = \left|\frac{x}{R}\right|^n \leq 1 \quad (\forall)\, n \geq 0, (\forall)\, x \in [0, R]$

$|f_n(x)| = \left(\frac{x}{R}\right)^n$

$0 \leq \frac{x}{R} \leq 1 \Rightarrow \underbrace{\left(\frac{x}{R}\right)^{n+1}}_{f_{n+1}''(x)} \leq \underbrace{\left(\frac{x}{R}\right)^n}_{f_n(x)}$

Se aplică ABEL

(i) și (ii)

$S: (-R, R] \to \mathbb{C}$

$S(x) = \sum_{n=0}^{\infty} a_n x^n$, $x \in (-R, R]$

din i) $\Rightarrow$ S e continuă pe $[0, R]$

(în R) $S(R) = \lim_{\substack{x \to R \\ x < R}} S(x)$

Din i) $\Rightarrow \sum_{n \geq 0} \left(\int_0^R a_n x^n dx \right)$ este conv. și $\int_0^R S(x)\,dx = \sum_{n=0}^{\infty} \left(\int_0^R a_n x^n \right)_0$

(T) $\sum_{n \geq 0} a_n x^n$, R - raza de conv

Pp. că $0 < R < \infty$.

Pp. că $\sum_{n \geq 0} a_n (R)^n$ e conv.

i) $\sum_{n \geq 0} a_n x^n$ e uniform conv. pe $[-R, 0]$

ii) $\sum_{n=0}^{\infty} a_n (-R)^n = \lim_{\substack{x \to -R \\ x > -R}} \left(\sum_{n=0}^{\infty} a_n x^n \right)$

iii) $\int_0^{-R} \left(\sum_{n=0}^{\infty} a_n x^n \right) dx = \sum_{n=0}^{\infty} a_n \frac{(-R)^{n+1}}{n+1}$

Teorema (Abel): Pp. că $\sum_{n \geq 0} a_n$ este conv. ($a_n \in \mathbb{C}$)

Atunci raza de conv. a seriei $\sum_{n \geq 0} a_n x^n$ este $R \geq 1$ și

$\sum_{n=0}^{\infty} a_n = \lim_{\substack{x \to 1 \\ x < 1}} \sum_{n=0}^{\infty} (a_n x^n)$

Dem: ~~Fie~~

Prop: $\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$, $(\forall)\ x \in (-1, 1]$

(pt. $x=-1$ seria e div.)

(seria are $R=1$)

$\ln 2 = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} \dots$

Dem: $[\ln(1+x)]' = \frac{1}{1+t} = \frac{1}{1-(-t)} = \sum_{n=0}^{\infty} (-t)^n = \sum_{n=0}^{\infty} (-1)^n t^n$

$t \in (-1,1)$, $|t|<1$

pt $|t| \geq 1$ $\Rightarrow \sum_{n\geq 0} (-1)^n t^n$ Div.

- pt. $|t|<1 \Rightarrow \sum_{n\geq 0} (-1)^n t^n$ conv.
- pt $|t| \geq 1 \Rightarrow \sum_{n\geq 0} (-1)^n t^n$ div

(serie geo. de ratie $(-t)$)

$\Rightarrow R$ a seriei $\sum_{n\geq 0} (-1)^n t^n$ este $R=1$

$\Rightarrow \sum_{n\geq 0} (-1)^n \frac{t^{n+1}}{n+1}$ are acelasi raza de conv, $R=1$

$\sum_{k\geq 0} (-1)^{k-1} \frac{t^k}{k}$

$\sum_{n=0}^{\infty} (-1)^n t^n = \frac{1}{1+t}$, $(\forall)\ t \in (-1,1)$

$-R'' \quad R''$

Fie $x \in (-1,1)$, $x \neq 0$, $|x|<1$

integrăm termen cu termen pe $[0, x]$ sau $[x, 0]$

$\int_0^x \frac{1}{1+t}\,dt = \sum_{0}^{\infty} (-1)^n \frac{t^{n+1}}{n+1}\Big|_0^x$

$\ln(1+t)\Big|_0^x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}$ $(\forall)\ x \in (-1,1)$, $x \neq 0$

evidentă și pt. $x=0$

$x=-1$: $\sum_{n\geq 1} \frac{-1}{n}$ Div ($\sum \frac{1}{n}$ div)

$x=1$: $\sum_{n\geq 1} \frac{1}{n}$ conv (Leibniz)

$$\Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \lim_{\substack{x\to 1 \\ x<1}} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n = \lim_{\substack{x\to 1 \\ x<1}} \ln(x+1)$$

[illegible]

Prop $\operatorname{arctg} x = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{2n+1}}{2n+1}$ $(\forall)\ x \in [-1,1]$

raza de conv. este $R=1$

$$\frac{\pi}{4} = \operatorname{arctg} 1 = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \dots$$

$\frac{1}{1+t^2}$ pt $(t)<1<1 \iff 1-t^2 < 1$

$|t|^2$

$$\frac{1}{1+t^2} = \sum_{n=0}^{\infty} (-t^2)^n = \sum_{n=0}^{\infty} (-1)^n \cdot t^{2n}$$

pt. $|t|<1 \Rightarrow \sum_{n\geq 0} (-1)^n t^{2n}$ Conv

pt $|t|\geq 1 \Rightarrow \sum_{n\geq 0} (-1)^n t^{2n}$ div

$\Rightarrow \sum_{n\geq 0} (-1)^n t^{2n}$ are raza de conv $R=1$

seria $\sum_{n\geq 0} (-1)^n \frac{t^{2n+1}}{2n+1}$ are aceeași rază ca

$|t|<1$

$\frac{1}{1+t^2} = \sum_{n=0}^{\infty} (-1)^n \cdot t^{2n}$, $R=1$

$|x|<1$, $x \neq 0$, $[0,x] \subseteq (-R,R)$
$[x,0]$

$\int_0^x \frac{1}{1+t^2}\,dx = \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n+1}}{2n+1}\Big|_0^x$

$\arctg t\Big|_0^x = \sum_{n=0}^{\infty} (-1)^n x^{2n+1}$

$\arctg x$ ////

$(\forall)\ x \in (-1,1)$
(o nu i;)

$x=1$: $\sum_{n\geq 0} \frac{(-1)^n}{2n+1}$ conv. (LEIBNIZ)

$\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \lim_{\substack{x\to 1 \\ x<1}} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = \arctg 1 = \frac{\pi}{4}$

$\arctg x$

pt $x=-1$:

$\sum_{n=0}^{\infty} \frac{(-1)^n \cdot (-1)}{2n+1} = -\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = -\arctg 1 = +\arctg(-1)$

Prop Seria Binomială

Fie $\alpha \in \mathbb{R} \setminus \mathbb{N}$ =>

$$(1+x)^\alpha = 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1)(\alpha-2)\cdot\ldots\cdot(\alpha-n+1)}{n!} \cdot x^n, \ (\forall)\ x \in (-1,1)$$

$R = 1$

① Lui MERTENS

$(u_n)_{n\geq 0}$ și $(v_n)_{n\geq 0}$ șiruri de nr. complexe

Pp că $\sum_{n\geq 0} u_n$ este abs. conv. și $\sum_{n\geq 0} v_n$ este conv.

$$c_n = u_0 v_n + u_1 v_{n-1} + \ldots + u_n v_0 = \sum_{k=0}^{n} u_k v_{n-k} = \sum_{k=0}^{n} u_{n-k} v_k$$

Atunci: $\sum_{n\geq 0} c_n$ este conv. și $\left(\sum_{n=0}^{\infty} u_n\right)\cdot\left(\sum_{n=0}^{\infty} v_n\right) = \sum_{n=0}^{\infty} c_n$

Obligatoriu măcar una să fie absolut conv.

Funcția exponențială

Prop: R seriei $\sum_{n\geq 0} \frac{z^n}{n!}$ este $R = \infty$

Fie $E(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!}$, $(\forall)\ z \in \mathbb{C}$

Af: $(\forall)\ z_1, z_2 \in \mathbb{C}$ => $\boxed{E(z_1)\cdot E(z_2) = E(z_1+z_2)}$

Dem: $a_n = \frac{1}{n!}$; $\frac{|a_{n+1}|}{|a_n|} = \frac{n!}{(n+1)!} = \frac{1}{n+1} \longrightarrow 0 = l$

$R = \frac{1}{l} = \infty$

$(\forall)\ z \in \mathbb{C} \Rightarrow \sum_{n\geq 0} \frac{z^n}{n!}$ este abs. conv.

$|z| < R = \infty$

$\sum_{n\geq 0} \frac{z_1^n}{n!} \cdot \sum_{n\geq 0} \frac{z_2^n}{n!}$ / amb. abs. conv.

$u_n = \frac{z_1^n}{n!}$; $v_n = \frac{z_2^n}{n!}$ $\xrightarrow{\text{Mertens}}$

$$c_n = \sum_{k=0}^{n} u_{n-k} v_k = \sum_{k=0}^{n} \left(\frac{z_1^{n-k}}{(n-k)!} \cdot \frac{z_2^k}{k!} \cdot n! \right) \cdot \frac{1}{n!} =$$

$$= \frac{1}{n!} \sum_{k=0}^{n} C_n^k z_1^{n-k} \cdot z_2^k = \frac{1}{n!} (z_1 + z_2)^n$$

$$\left(\sum_{n=0}^{\infty} u_n \right) \left(\sum_{n=0}^{\infty} v_n \right) = \sum_{n=0}^{\infty} c_n$$

$$\left(\sum_{n=0}^{\infty} \frac{z_1^n}{n!} \right) \left(\sum_{n=0}^{\infty} \frac{z_2^n}{n!} \right) =$$

$$E(z_1) \cdot E(z_2) = E(z_1 + z_2)$$

Prop:

i) $E(0) = 1$

ii) $(\forall)\ z \in \mathbb{C} \Rightarrow E(z) \neq 0$ și $E(-z) \frac{1}{E(z)}$

iii) $(\forall)\ n \in \mathbb{N}$, $(\forall)\ z \in \mathbb{C} \Rightarrow E(nz) = (E(z))^n$

---

i) $S_n(z) = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \ldots + \frac{z^n}{n!}$

$S_n(0) = 1 \longrightarrow 1$

$\searrow E(0)$ $\quad 1 + \sum_{n=1}^{\infty} \frac{0^n}{n!} = 1$

ii) $E(0) = 1 = E\big(z + (-z)\big) = E(z) \cdot E(-z)$

$E(z_1) \cdot E(z_2) \cdots E(z_n) = E(z_1 + z_2 + \dots + z_n)$

$z_1 = z_2 = \dots = z_n = z$

$\big(E(z)\big)^n = E(nz)$.

Obs: $(\forall)\ x \in \mathbb{R} \Rightarrow E(x) \in \mathbb{R}$

Prop a) Fie $v \in \mathbb{C}$. Fie $\varphi : \mathbb{R} \to \mathbb{C}$, $\varphi(x) = E(vx) = \sum_{n=0}^{\infty} \frac{v^n x^n}{n!}$

A1: φ e derivabilă și $\varphi'(x) = v\, E(vx)$

adică $\varphi'(x) = v \cdot \varphi(x)$

b) Fie $\psi : \mathbb{R} \to \mathbb{R}$, $\psi(x) = E(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

A1. ψ e derivabilă și $\psi'(x) = \psi(x)$

și $\psi(0) = 1$

$E'(x) = E(x)$, $(\forall) x \in \mathbb{R}$

$E(0) = 1$

Dem $\sum_{n \geq 0} \frac{v^n x^n}{n!}$ e convergentă $(\forall) x \in \mathbb{R}$ $(z = v \cdot x)$

$\Rightarrow R = \infty$

$(-R, R) = (-\infty, \infty)$

$\varphi(x) = \sum_{n=0}^{\infty} \frac{v^n x^n}{n!}$, $(\forall) x \in \mathbb{R}$

e indefinit derivabilă pe $(-\infty, \infty)$ și $\varphi'(x) = \sum_{n=1}^{\infty} \frac{v^n n x^{n-1}}{n!} =$

$= v \sum_{n=1}^{\infty} \frac{v^{n-1} x^{n-1}}{(n-1)!} = v \sum_{k=0}^{\infty} \frac{v^k x^k}{k!} = v \cdot \varphi(x)$

$\quad\quad\quad

Prop: Fie $f: \mathbb{R} \to \mathbb{R}$, derivabilă

Dacă:

$f'(x) = f(x), (\forall) x \in \mathbb{R}$, $f(0) = 1 \Rightarrow$

$\Rightarrow f(x) = E(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, $(\forall) x \in \mathbb{R}$

Dem: Fie $g: \mathbb{R} \to \mathbb{R}$

$g(x) = f(x) \cdot E(-x)$

$g'(x) = \underbrace{f'(x)}_{f(x)} \cdot E(-x) + f(x) \cdot \underbrace{E'(-x)}_{E(-x)} \cdot (-1)$

$g'(x) = f(x) E(-x) - f(x) E(x) = 0 \quad (\forall) x \in \mathbb{R}$

$\Rightarrow (\exists) c \in \mathbb{R}$ a.î. $g(x) = c, \quad (\forall) x \in \mathbb{R}$

$f(x) \cdot E(-x) = c$

$x = 0: \quad \underset{1}{f(0)} \cdot \underset{1}{E(0)} = c = 1$

$f(x) \cdot \frac{1}{E(x)} = 1 \Rightarrow f(x) = E(x)$

Obs: $E(x) > 0, (\forall) x \in \mathbb{R}$

$E(x) = E\left(\frac{x}{2}, \frac{x}{2}\right) = \left(E\left(\frac{x}{2}\right)\right)^2 \geq 0$

$E(x) \neq 0 \Big/ \Rightarrow E(x) > 0$

Fie $e = E(1) = \sum_{n=0}^{\infty} \frac{1}{n!}$

$E(n \cdot 1) = (E(1))^n$, $E(n) = e^n$, $(\forall)\ n \in \mathbb{N}$

$E(-n) = \frac{1}{E(n)} = \frac{1}{e^n} = e^{-n}$ $(\forall)\ n \in \mathbb{N}$

$E(n) = e^n$, $(\forall)\ n \in \mathbb{Z}$

$q > 0$, $q \in \mathbb{N}$

$E\left(q \cdot \frac{1}{q}\right) = \left(E\left(\frac{1}{q}\right)\right)^q = e$ (since $E(1) = e$, $E\left(\frac{1}{q}\right) > 0$)

$\Rightarrow E\left(\frac{1}{q}\right) = \sqrt[q]{e} = e^{\frac{1}{q}}$

$r = \frac{p}{q} \in \mathbb{Q}$, $r > 0$, $p, q \in \mathbb{N}$, $q \neq 0$

$E(r) = E\left(p \cdot \frac{1}{q}\right) = \left(E\left(\frac{1}{q}\right)\right)^p = \left(\sqrt[q]{e}\right)^p = \sqrt[q]{e^p} = e^{\frac{p}{q}} = e^r$

$E(r) = e^r$, $(\forall)\ r \in \mathbb{Q}$, $r > 0$

$E(0) = e^0 = 1$

$E(-r) = \frac{1}{E(r)} = \frac{1}{e^r} = e^{-r}$

Fie $x \in \mathbb{R} \setminus \mathbb{Q}$, $\exists (r_n)_n \subset \mathbb{Q}$, $r_n \to x$

$e^x = \lim_{n \to \infty} e^{r_n} = \lim_{n \to \infty} E(r_n) = E(x) = e^x$ (continuă)

$E(x) = e^x$, $(\forall)\ x \in \mathbb{R}$

$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, $(\forall)\ x \in \mathbb{R}$

Se prelungește funct. exp. la $\mathbb{C}$

$$e^z = E(z), \ (\forall) z \in \mathbb{C}$$

$$e^z \overset{\text{def}}{=} \sum_{n=0}^{\infty} \frac{z^n}{n!}, \ (\forall) z \in \mathbb{C}$$

Prop: 1) $e^{z_1} \cdot e^{z_2} = e^{z_1+z_2}$, $(\forall) z_1, z_2 \in \mathbb{C}$

2) $(\forall) z \in \mathbb{C} \Rightarrow e^z \neq 0$ și $e^{-z} = \frac{1}{e^z}$

3) $e^{nz} = (e^z)^n$, $(\forall) n \in \mathbb{N}$, $(\forall) z \in \mathbb{C}$

Prop: Fie $v \in \mathbb{C}$

$\varphi: \mathbb{R} \to \mathbb{C}$, $\varphi(x) = e^{vx}$, $(\forall) x \in \mathbb{R}$

φ e derivabilă și $\varphi'(x) = v \cdot e^{vx}$

$(\forall) x \in \mathbb{R} \Rightarrow \varphi'(x) = v\varphi(x)$)

(T) $\cos x = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{2n}}{(2n)!}$

$\sin x = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{2n+1}}{(2n+1)!}$ $\Big/ R = \infty$

Serii Leibniz

$(a_n)_{n \geq 0}$ șir de nr. Reale. Pp. că $a_n \to 0$

Pp că $(\exists) N \geq 0$, $N \in \mathbb{N}$, a.î. $(\forall) n \geq N$

avem $a_{n+1} \leq a_n$

a) Atunci $\sum_{n \geq 0} (-1)^n a_n$ (resp. $\sum_{n \geq 0} (-1)^{n-1} a_n$) este conv

Fie $S_n = (-1)^2 a_2 + \dots + (-1)^n \cdot a_n$, $n \geq 2$

(resp. $S_n = (-1)^{2-1} a_2 + \dots + (-1)^{n-1} a_n$)

Fie S - suma seriei:

$S = \sum_{n=2}^{\infty} (-1)^n a_n$ (resp. $S = \sum_{n=2}^{\infty} (-1)^{n-1} a_n$)

$S_n \to S$

B) $(\forall) n \geq N \Rightarrow |S_n - S| \leq a_{n+1}$

C) i) pt. $2n \geq N \Rightarrow S_{2n+2} \leq S_{2n}$ (resp $S_{2n+2} \geq S_{2n}$)

ii) pt $2n-1 \geq N \Rightarrow S_{2n+1} \geq S_{2n-1}$ (resp $S_{2n+1} \leq S_{2n-1}$)

iii) $(\forall) n \geq N$, npar $\Rightarrow S_n \geq S$ (resp $S_n \leq S$)

iv) $(\forall) n \geq N$, nimpar $\Rightarrow S_n \leq S$ (resp $S_n \geq S$)

d) Dacă $n \geq N$ și S_n are forma

$S_n = \dots (+) a_n \Rightarrow S_n \geq S$

Dacă $S_n = \dots (-) a_n \Rightarrow S_n \leq S$

$S_n \leq S \quad |S_n - S| < \varepsilon$

$S_n \leq S \leq S_n + \varepsilon$

2 zecimale $\varepsilon = ?$

???

Ex: Calc $\int_0^1 e^{-x^2} dx$ cu 2 zecimale exact

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad (\forall)\ x \in \mathbb{R}, \quad R = \infty$$

$$e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!}, \quad (\forall)\ x \in \mathbb{R}$$

$$R = \infty$$

$$[a,b] \subseteq (-R, R)$$

integr. termen cu termen.

$$\int_0^1 e^{-x^2} dx = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{n!(2n+1)} \Big|_0^1$$

$$\int_0^1 e^{-x^2} dx = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{1}{n!(2n+1)} = a_n, \quad a_n \to 0$$

S Leibniz

$$\frac{a_{n+1}}{a_n} = \frac{1}{n+1} \cdot \frac{2n+1}{2n+3} < 1, \quad a_{n+1} < a_n, \ (\forall)\ n \geq 0$$

$$S_n = a_0 - a_1 + \ldots + (-1)^n a_n \longrightarrow S$$

$$|S_n - S| \leq a_{n+1}, \ (\forall)\ n \geq 0$$

$$|S_{n-1} - S| \leq a_n, \ (\forall)\ n \geq 1$$

??n?? $a_n = 0,00\ldots$

$$a_n = \frac{1}{n!(2n+1)}$$

$a_0 = 1, \quad a_1 = \frac{1}{3} > \frac{1}{10} = 0,1$

$a_2 = \frac{1}{10} = 0,1$

$a_3 = \frac{1}{6 \cdot 7} = \frac{1}{42} > \frac{1}{50} = 0,02$

$a_4 = \frac{1}{4! \cdot 9} = \frac{1}{216} < \frac{1}{200} = 0,005$

$n = 4$

$|S_3 - S| \leq a_4 \leq 0,005$

$S_3 = a_0 - a_1 + a_2 - a_3 \ /\Rightarrow$

$\Rightarrow S - S_3 \leq 0,005$

$S_3 \leq S \leq S_3 + 0,005$

$S_3 = 1 + \frac{1}{10} - \frac{1}{3} - \frac{1}{42} = \frac{11}{10} - \frac{15}{42} = \frac{11}{10} - \frac{15}{42} = \frac{11}{10} - \frac{5}{14} = \frac{52}{70} = 0,742$

$\Rightarrow S_3 = 0,742\ldots$

$S_3 \le S \le S_3 + 0,005 = 0,747\ldots \Rightarrow$

$0,742 \le S \le 0,747 \Rightarrow S = \int_0^1 e^{-x^2} dx = 0,74\ldots$

Ex: Calc cu 4 zec mode exacte:

$\int_0^1 \frac{\sin(x^2)}{x^2} dx$

$\lim_{x\to 0} \frac{\sin x^2}{x^2} = 1$; $f(x) = \begin{cases} \frac{\sin x^2}{x^2}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

l cont.

$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$, $(\forall)\ x \in \mathbb{R}$

$R = \infty$

$\sin x^2 = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{4n+2}}{(2n+1)!} \Big/ \cdot \frac{1}{x^2}$

$\frac{\sin x^2}{x^2} = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{4n}}{(2n+1)!}$, $(\forall)\ x \in \mathbb{R}^*$

$R = \infty$

$[0,1] \subseteq (-R, R) = (-\infty, \infty)$

$= a_n$

$\int_0^1 \frac{\sin x^2}{x^2} dx = \sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+1}}{(2n+1)!(4n+1)} \Big|_0^1 = \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)!(4n+1)}$

$S = \sum_{n=0}^{\infty} (-1)^n \cdot a_n$ $\quad a_n \to 0$

$\frac{a_{n+1}}{a_n} = \frac{1}{(2n+2)(2n+3)} \cdot \frac{4n+1}{4n+5} < 1 \Rightarrow a_{n+1} < a_n$

$S_n = a_0 - a_1 + \dots + (-1)^n a_n \to S$

$|S_n - S| \le a_{n+1}, (\forall) n \ge 0$

$|S_{n-1} - S| \le a_n, (\forall) n \ge 1$

? n ?

$a_0 = 1$

$a_1 = \frac{1}{3! \cdot 5} = \frac{1}{30}$

$a_2 = \frac{1}{5! \cdot 9} > \frac{1}{120 \cdot 9} > \frac{1}{1080} > \frac{1}{1000} = 0,001$

$a_3 = \frac{1}{7! \cdot 13} = \frac{1}{120 \cdot 6 \cdot 7 \cdot 13} = \frac{1}{5040 \cdot 13} < \frac{1}{50000} = 0,00002$

pt $n = 3 \Rightarrow$

$|S_2 - S| \le a_3 < 2 \cdot 10^{-5}$

$S_2 = a_0 - a_1 + a_2 \ge S$

$S_2 - S < 2 \cdot 10^{-5}$

$S_2 - 2 \cdot 10^{-5} \le S \le S_2$

$S_2 = \frac{1045}{1080} = 0,96759\dots$

$0,96757 \le S \le S_2 = 0,96759\dots$

$S = \int_0^1 \frac{\sin(x^2)}{x^2} dx = 0,9675$

02.11	AMat
1	GHiu

Ec. cos și sin.

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

$$e^{iz} = \sum_{n=0}^{\infty} \frac{i^n z^n}{n!}$$

$$e^{-iz} = \sum_{n=0}^{\infty} \frac{(-i)^n z^n}{n!}$$

$$e^{iz} + e^{-iz} = \sum_{k=0}^{\infty} \underbrace{\frac{i^k + (-i)^k}{k!}}_{=v_k} z^k$$

$k = 2n$

$$v_{2n} = \frac{(i^2)^n + (i^2)^n}{(2n)!} = 2\frac{(-1)^n}{(2n)!}$$

$k = 2n+1$

$$v_{2n+1} = \frac{i^{2n+1} - i^{2n+1}}{(2n+1)!} = 0$$

$$\frac{e^{iz} + e^{-iz}}{2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n!} \cdot z^{2n} \quad (\forall) z \in \mathbb{C}$$

$$e^{iz} - e^{-iz} = \sum_{k=0}^{\infty} \underbrace{\frac{i^k - (-i)^k}{k!}}_{w_k} \cdot z^k, \quad (\forall) z \in \mathbb{C}$$

$k = 2n$

$$w_{2n} = \frac{i^{2n} - i^{2n}}{2n!} = 0$$

$k = 2n+1$

$$w_{2n+1} = \frac{i \cdot (-1)^n + i \cdot (-1)^n}{(2n+1)!} = \frac{2i(-1)^n}{(2n+1)!}$$

$$\frac{e^{iz} - e^{-iz}}{2i} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1}, \quad (\forall)\ z \in \mathbb{C}$$

$$C(z) = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{z^{2n}}{(2n)!}, \quad (\forall)\ z \in \mathbb{C}$$

are rază de conv. $R = \infty$

$$S(z) = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{z^{2n+1}}{(2n+1)!}, \quad (\forall)\ z \in \mathbb{C}$$

are $R = \infty$

$$C(z) = \frac{e^{iz} + e^{-iz}}{2} \; ; \; S(z) = \frac{e^{iz} - e^{-iz}}{2i}$$

$$C(-z) = \frac{e^{-iz} + e^{iz}}{2} = C(z) \quad \text{pară}$$

$$S(-z) = \frac{e^{-iz} - e^{iz}}{2i} = -S(z) \quad \text{impară}$$

$$C(0) = \frac{e^0 + e^0}{2} = 1 \; ; \; S(0) = \frac{e^0 - e^0}{2i} = 0$$

$$\underline{C(0) = 1 \; ; \; S(0) = 0}$$

$$C(z) + i\,S(z) = \frac{e^{iz} + e^{-iz}}{2} + \frac{e^{iz} - e^{-iz}}{2}$$

$$e^{iz} = C(z) + i\,S(z)$$

$$e^{-iz} = C(z) + iS(-z) = C(z) - iS(z)$$

$$1 = e^{iz} \cdot e^{-iz} = (C(z))^2 + (S(z))^2$$

$$(C(z))^2 + (S(z))^2 = 1$$

$x \in \mathbb{R}$

$$C(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \in \mathbb{R}$$

$$S(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \in \mathbb{R}$$

$$C(x) = \frac{e^{ix} + e^{-ix}}{2} \quad ; \quad S(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

$$C'(x) = \frac{1}{2}\left(ie^{ix} - ie^{-ix}\right) = \frac{-i}{2} \cdot (-1)\left(e^{ix} - e^{-ix}\right) =$$

$$= \frac{i}{2} \cdot 2i \cdot S(x) = -S(x)$$

$$S'(x) = \frac{ie^{ix} + ie^{-ix}}{2i} = C(x)$$

$C'(x) = -S(x)$; $S'(x) = C(x)$

$C(0) = 1$; $S(0) = 0$

Prop. Fie $f, g : \mathbb{R} \to \mathbb{R}$ derivabile a.î.

$(\forall)\ x \in \mathbb{R}$ avem: $f'(x) = -g(x)$ și $g'(x) = f(x)$

Pp că $f(0) = 1$; $g(0) = 0$

Atunci:

$$f(x) = C(x) = \sum_{k=0}^{\infty} (-1)^k \cdot \frac{x^{2k}}{(2k)!}$$

$$g(x) = S(x) = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{2n+1}}{(2n+1)!}$$

Demo: Fie $h : \mathbb{R} \to \mathbb{C}$; $h(x) = (f(x) + ig(x)) \cdot e^{-ix}$

$$h'(x) = (\underbrace{f'(x)}_{-g(x)} + i\underbrace{g'(x)}_{f(x)})\, e^{-ix} - ie^{-ix}(f(x) + ig(x)) =$$

$$= (-g(x) + if(x))\, e^{-ix} - e^{-ix}(if(x) - g(x)) = 0 \Rightarrow (\exists)\, c \in \mathbb{C} \text{ a.î. } h(x) = c$$

$(f(x) + i g(x))\, e^{-ix} = C, \quad \forall x \in \mathbb{R}$

$x = 0$

$(f(0) + i g(0)) \cdot 1 = C \Rightarrow C = 1$

$f(x) + i g(x) = e^{ix} = C(x) + i S(x)$

$\cos' = -\sin ; \quad \sin' = \cos$

$\cos(0) = 1 ; \quad \sin(0) = 0$

$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$; $\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$

$\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}$; $\sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}$

$\cos z = \frac{e^{iz} + e^{-iz}}{2}$; $\sin z = \frac{e^{iz} - e^{-iz}}{2i}$

$\cos(-z) = \cos z$; $\sin(-z) = -\sin(z)$

$\cos 0 = 1 ; \quad \sin 0 = 0$

$e^{iz} = \cos z + i \sin z$

$\cos^2 z + \sin^2 z = 1$

$e^{ix} = \cos x + i \sin x$

$\cos x = \mathrm{Re}(e^{ix})$; $\sin x = \mathrm{Im}(e^{ix}), \quad \forall x \in \mathbb{R}$

$e^z = e^{x+iy} = e^x (\cos y + i \sin y)$

$e^{\mathrm{Re}\, z} = |e^z|$

$e^{i\frac{\pi}{2}} = \cos \frac{\pi}{2} + (\sin \frac{\pi}{2}) i = i$

(0 and 1)

$$\left(e^{i\frac{\pi}{2}}\right)^2 = i^2 = -1$$

$$\| \quad e^{i\pi}$$

$$e^{i\pi} + 1 = 0$$

$$e^{i\pi} = -1$$

$$e^{z + 2k\pi i} = e^z \cdot e^{2k\pi i}$$

$$\| \quad 1$$

$$e^{z + 2k\pi i} = e^z$$

$$\cos(a+b) = \cos a \cos b - \sin a \sin b \quad (\forall)\ a, b \in \mathbb{C}$$

$$\cos(a+b) = \frac{e^{i(a+b)} + e^{-i(a+b)}}{2}$$

$$\cos a \cos b = \frac{e^{ia} + e^{-ia}}{2} \cdot \frac{e^{ib} + e^{-ib}}{2}$$

$$\sin a \sin b = \frac{e^{ia} - e^{-ia}}{2i} \cdot \frac{e^{ib} - e^{-ib}}{2i}$$

$$ch\, z = \frac{e^z + e^{-z}}{2} \quad ; \quad sh\, z = \frac{e^z - e^{-z}}{2}$$

$$ch(-z) = ch\, z \quad ; \quad sh(-z) = -sh(z)$$

$$ch(iz) = \cos z \quad ; \quad sh(iz) = i \sin z$$

$$ch\, z = \cos(iz) \quad ; \quad sh\, z = -i \sin(iz)$$

$$\operatorname{ch} z = \cos iz = \sum_{n=0}^{\infty} (-1)^n \frac{(-1)^n \cdot z^{2n}}{(2n)!}$$

$$\operatorname{sh} z = (-i)\cdot \sin(iz) = -i \cdot i \sum_{n=0}^{\infty} \frac{(-1)^n (-1)^n z^{2n+1}}{(2n+1)!}$$

$$\operatorname{ch} z = \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!}$$

$$\operatorname{sh} z = \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!}$$

$$\operatorname{ch} x = \frac{e^x + e^{-x}}{2} \qquad \operatorname{sh} x = \frac{e^x - e^{-x}}{2}$$

$$\operatorname{ch}' x = \frac{e^x - e^{-x}}{2} \qquad \operatorname{sh}' x = \frac{e^x + e^{-x}}{2}$$

$\cos z = \operatorname{ch}(iz)$; $i \operatorname{sh} z = \sin iz$

$\operatorname{ch}'(x) = \operatorname{sh}(x)$; $\operatorname{sh}'(x) = \operatorname{ch}$

$\operatorname{ch} 0 = 1$; $\operatorname{sh} 0 = 0$

$\cos^2 iz + \sin^2 iz = 1$

$\operatorname{ch}^2 z + i^2 \operatorname{sh}^2 z = 1$

$(\operatorname{ch} z)^2 - (\operatorname{sh} z)^2 = 1$

F. lui Taylor

(T) ROLLE : Fie $f: [a,b] \longrightarrow \mathbb{R}$

f cont pe $[a,b]$; f derv. pe (a,b)

Daca $f(a) = f(b)$; at. $(\exists)\ c \in (a,b)$ a.i. $f'(c) = 0$

(T) CAUCHY : Fie $f, g: [a,b] \longrightarrow \mathbb{R}$

continue pe $[a,b]$, derivabile pe (a,b).

Pp. că $(\forall)\ x \in (a,b)$, avem $g'(x) \neq 0$.

Atunci i) $g(a) \neq g(b)$

ii) $(\exists)\ c \in (a,b)$ a.i. $\dfrac{f(b)-f(a)}{g(b)-g(a)} = \dfrac{f'(c)}{g'(c)}$

Dem: i) Pp $g(a) = g(b)$ $\overset{\text{ROLLE}}{\Rightarrow}$ $(\exists)\ k \in (a,b)$ a.î. $g'(k) = 0$

Fals

ii)

Fie $h(x) = (f(b) - f(a))\,g(x) - (g(b) - g(a))\,f(x)$

Probăm că $h(a) = h(b)$ și apl. (T) ROLLE

Corolar

Dacă în T. CAUCHY

se ia $g(x) = x$, $g'(x) = 1 \neq 0$

se obț. T. LAGRANGE

De Enunțat!

Teorem: F. lui TAYLOR

Fie $I \subseteq \mathbb{R}$, I interval

$f: I \to \mathbb{R}$ de $n+1$ ori derivabilă

Fie $a \in I$, Atunci: $(\forall)\ x \in I$, $x \neq a$ $(\exists)\ c \in (a,x)$ sau $c \in (x,a)$

a.î. $f(x) = \underbrace{\sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k}_{\text{pol. nul.}} + \underbrace{\frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}}_{\text{restul sub forma lui LAGRANGE}}$

$T_{n,a}(x)$ - s.n. polinomul Taylor de ordinul n

$f^{(0)}(x) = f(x)$

$C_{n,a,x}$

$F(t) = \sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(x-t)^k$; $G(t) = (x-t)^{n+1}$

Aplicații T. CAUCHY pe $[a, x]$ sau $[x, a]$

Corolar

Fie $P(x) \in \mathbb{R}[X]$, grad $P[X] = n$

Dem. $P(x) = \sum_{k=0}^{n} \frac{P^{(k)}(a)}{k!}(x-a)^k$

Ex: Calc. $I = \int_0^{1/3} \frac{\ln(1-x)}{x}\,dx$ cu 2 zecimale exacte

$\lim_{x\to 0} \frac{\ln(1-x)}{x} = \lim_{x\to 0} \frac{-\ln(1-x)}{-x} = -1$

$\varphi(x) = \begin{cases} \frac{\ln(1-x)}{x}, & x < 1, x \neq 0 \\ -1, & x = 0 \end{cases}$

φ e cont.

$I = \int_0^{1/3} \varphi(x)\,dx$

Fie $f: (-\infty, 1) \to \mathbb{R}$, $f(x) = \ln(1-x)$, $\forall x < 1$

$a = 0$

Fie $x \in (0, \frac{1}{3}]$, $(\exists)\, c \in (0, x)$ a.î. $f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!}x^k +$

$+ \frac{f^{(n+1)}(c)}{(n+1)!}x^{n+1}$

$f(x) = \ln(1-x)$; $f'(x) = \frac{-1}{1-x}$; $f''(x) = (-1)\cdot\frac{(+1)}{(1-x)^2}$

$f'''(x) = -\frac{1\cdot(-2)}{(1-x)^3}$ $\qquad f(0) = \ln 1 = 0$

$f^{(k)}(x) = -\frac{(k-1)!}{(1-x)^k}$

↑ Pp adevărată

$f^{(k+1)}(x) = -\frac{(k-1)!\,(-k)}{(1-x)^{k+1}} = \frac{-k!}{(1-x)^{k+1}}$

$\frac{f^{(k)}(x)}{k!} = -\frac{1}{k(1-x)^k}$; $\frac{f^{k}(0)}{k!} = \frac{-1}{k}$

$n \geq 1$

$0 < x \leq \frac{1}{3}$; $0 < c < x$

$f(x) = \ln(1-x) = \sum_{k=1}^{n} \frac{-1}{k} x^k + \frac{-x^{n+1}}{(n+1)(1-c)^{n+1}}$

$-f(x) - \sum_{k=1}^{n} \frac{x^k}{k} = \frac{x^{n+1}}{(n+1)(1-c)^{n+1}}$

$0 < c < x \leq \frac{1}{3}$

$-c \geq -\frac{1}{3} \;/+1$; $1-c \geq \frac{2}{3}$

$(1-c)^{n+1} \geq \frac{2^{n+1}}{3^{n+1}}$; $\frac{1}{(1-c)^{n+1}} \leq \frac{3^{n+1}}{2^{n+1}}$

$0 < \frac{x^{n+1}}{(n+1)(1-c)^{n+1}} \leq \frac{x^{n+1}}{(n+1)} \cdot \frac{3^{n+1}}{2^{n+1}}$

$-f(x) = -\sum_{k=1}^{n} \frac{x^k}{k}$

$$0 \le -\underbrace{\frac{f(x)}{x}}_{\varphi(x)} - \sum_{k=1}^{4} \frac{x^{k-1}}{k} \le \frac{x^4}{4+1} \cdot \frac{3^{4+1}}{2^{4+1}}$$

~~$0 \le x \le \frac{1}{3}$~~ $0 < x \le \frac{1}{3}$

pt. $x=0$

$$0 \le -(-1) - \frac{1}{1} \le 0 \iff 0 = 0 = 0$$

$$0 \le -\underbrace{\int_0^{1/3} \varphi(x)\,dx}_{=J} - \sum_{k=1}^{4} \frac{x^k}{k^2}\Big|_0^{1/3} \le \frac{3^{4+1}}{(4+1)^2 2^{4+1}} \cdot x^{4+1}\Big|_0^{1/3}$$

$$0 \le -J - \sum_{k=1}^{4} \frac{1}{k^2 \cdot 3^k} \le \underbrace{\frac{3^{4+1}}{(4+1)^2 \cdot 2^{4+1}} \cdot \frac{1}{3^{4+1}}}_{= \frac{1}{(4+1)^2 \cdot 2^{4+1}} = \alpha_4}$$

$n=1$: $\frac{1}{16} = \alpha_1$

$n=2$: $\frac{1}{72} > \frac{1}{100}$

n_3: $\frac{1}{16 \cdot 16} = \frac{1}{256} < \frac{4}{1000} = 0{,}004$

$$0 \le -J - \sum_{k=1}^{3} \frac{1}{k^2 \cdot 3^k} \le \alpha_3 < 0{,}004$$

$$\sum_{k=1}^{3} \frac{1}{k^2 \cdot 3^k} \le -J \le \sum_{k=1}^{3} \frac{1}{k^2 \cdot 3^k} + 0{,}004$$

$$\frac{1}{3} + \frac{1}{4 \cdot 9} + \frac{1}{9 \cdot 3^3} = \frac{355}{972} = 0{,}365$$

$$\underbrace{\frac{355}{972}}_{0{,}365\ldots} \le -J \le \underbrace{\frac{355}{972} + 0{,}004}_{0{,}369}$$

$J = -0{,}36\ldots$

PARTIAL

TAYLOR inclusiv